

Equilibrium Provision of Information in Two-Sided Markets: Evidence from an Online Job Platform

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Abstract

This paper investigates the trade-off agents face when providing information to potential matches in two-sided markets. Specifically, we explore how, in online labor markets, market congestion can lead to under-provision of information from global workers to employers and ultimately negatively impact match efficiency. We develop a model of equilibrium information provision characterized by attention-constrained employers and workers who provide match-specific costly information to their potential employers. The model shows that congestion forms a binding attention constraint, leading to under-provision of signaling effort by workers in equilibrium. We then provide empirical evidence from a major international task-based online labor market that supports our theoretical model, showing that congestion leads to hasty and generic job applications and that the market is characterized by low job-filling rates. Our model and setting lay the groundwork needed to quantify the loss of efficiency that arises due to information under-provision in congested markets.

Keywords: Digital platforms, Two-sided matching markets, Trade in Services, Freelancing, Congestion, Communication Efforts

JEL Codes: C78, D47, J44, F23, L14

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1 Introduction

Finding a “match” in a two-sided market, whether it is an employer looking for a worker, a tenant looking for a house, or a student looking for a school, requires agents on at least one side to provide information about themselves to agents on the other side. While some of this information can be publicly observable, a key mechanism by which agents find high-value matches is by providing information about themselves that is specific to the potential match of interest. Providing such information can be costly. For example, in addition to submitting a generic CV, job searchers often spend time crafting a job-specific application tailored to a job post that signals they will be a good fit. These information provision costs present agents with a trade-off: Is it worth expending the effort to signal match-specific information about themselves, even when the match may not form?

In this paper, we study how agents navigate this information provision trade-off in equilibrium when competing for the same matches. In particular, we study how market congestion—when agents are overwhelmed with many potential matches, making it complicated to consider them all—can lead agents to limit their information provision in equilibrium, eroding match efficiency.

We explore these topics in the context of task-based online labor markets, an ideal and increasingly economically relevant setting ([Autor, 2001](#)) in which to study the information provision trade-off under congestion. Task-based online labor markets are particularly sensitive to the information provision trade-off. Because jobs listed on task-based online labor markets tend to be one-time isolated tasks, the cost of providing information (as an applicant) or of screening applicants (as an employer) is a larger fraction of the overall value of a hire relative to the in-person salaried labor market, thus raising the potential for information provision costs to impede efficient hiring. Yet information provision is essential to screening in these markets: because employers must commit to pay workers who they will never meet in person and will likely only have limited contact with, the onus is on applicants to provide information about themselves to their potential employers to convince them of their match quality with the job. Furthermore, in the presence of congestion, competition for each job increases, potentially leading workers to be less willing to spend time personalizing applications for each job. In equilibrium, all workers may limit their information provision by writing less specific job applications, eroding information transmission between workers and employers. The result is that employers may have low levels of trust in the applications they receive and leave their job posts unfilled, lowering even further the chances that workers costly provision of information will pay off.

To concretely establish the interaction between congestion and the information provision trade-off, we develop a model for equilibrium information provision between employers and workers on an online labor market in which employers are attention constrained and workers need to send match-specific costly information to convince employers to hire them. We build a static model of job matching in which many workers exogenously arrive at a job posting and send applications including a price bid and a signal of match quality. Employers then look through applications and choose a single worker or the outside option to maximize their expected payoffs.

Worker payoffs are only sensitive to their own opportunity costs of supplying their labor, the amount they will be paid, and the effort costs of signaling, while employers only care about the match quality of the workers they hire and the amount they pay them. While workers know their opportunity costs before applying, we make the crucial assumption that, conditional on observables, workers do not know their match value with the jobs for which they apply. The idea behind this assumption is that workers know exactly how costly it will be to supply their labor for a specific task, but they do not know exactly how the employers will value that labor once it has been completed. This informational assumption changes the model from one in which workers endeavor to share their private information about their type (Spence, 1973) to one in which workers choose to expend costly effort to send signals, the output of which will partially reveals their types (Holmström, 1999). This assumption allows for a tractable signaling model where workers only have one dimensional-private information – their costs – and employers do not form preferences over those private types. This part of the model encapsulates how workers can expend costly effort to provide information to employers.

A key aspect of our model is that employers are attention constrained. This is a prominent feature of the Freelancer.com platform, and thus matches our empirical setting.¹ They are not able to consider all of the applications they receive, and they rely on the platform to order applications for them according to a recommender system that only takes into account observable characteristics of workers, rather than their bids or signals. Workers do not know for sure whether their applications will be considered, but they form accurate beliefs over their chances of being considered and factor this into their information provision decisions. Thus, we show how congestion in the form of binding attention constraints leads to under provision of signaling effort in equilibrium and how this can lead the market to have inefficiently low job-filling rates.

¹Through our cooperation with Freelancer.com, we have access to “click data” which records which parts of the website employers and workers clicked on. The fact employers only consider a subset of the submitted job applications is a key feature of the platform, and a widely acknowledge fact among prospective freelancers discussing how to get jobs.

We then elucidate how our model is empirically tractable and sketch an argument for how the model could be identified with data from an online labor market.²

We use data from [Freelancer.com](https://www.freelancer.com), a major international online labor market, to present descriptive evidence of this hypothesized equilibrium in which congestion leads to under provision of information from workers to employers. On the employers’ side, we find that employers have to sort through many undifferentiated bids, many of which have been hastily written and poorly customized. These findings paired with the fact that, in our sample, only 40% of posted jobs get filled, suggests that workers are not providing enough information to convince employers to hire them, and as employers have to look through more and more of these insufficient applications, they opt for their outside options. On the workers’ side, we find that workers submit applications soon after projects are posted and measure that they spend very little time reading the job post and submitting their applications. These findings paired with the fact that, in our sample, only 10% of workers ever find jobs on the platform, and the acceptance rate per application is quite low for both workers with and without reputations on the platform lends suggestive evidence that workers internalize their low chances of consideration and accordingly spend little effort on applications. In sum, these findings are suggestive evidence for the forces highlighted in our model and suggest that future work should be done to quantify how much efficiency is lost to under-provision of information in equilibrium under congestion in this market.

Related Literature. Our paper makes three main contributions to the literature on online labor markets, empirical auctions, and information economics.

Online Labor Markets. First, we highlight a key tension in the design and maintenance of online labor markets, where market thickness can be a double-edged sword. Online labor markets have grown in recent years, globalizing traditionally local labor markets and enabling employers to outsource tasks to contractors worldwide ([Agrawal et al., 2015](#); [Kässi et al., 2021](#)). This growth has been accompanied by a growing body of literature on these markets. [Brinatti et al. \(2021\)](#) find that remote wages are sensitive to foreign competition and are higher for workers in regions with higher income per capita. [Horton et al. \(2021\)](#) study the role of cheap talk in market design and shows that signaling opportunities may help online labor markets reach more desirable equilibria. Another strand of literature explores the impact of market design features on matching outcomes in digital marketplaces. [Horton \(2017\)](#) and [Barach et al. \(2020\)](#) study the effects of algorithmic recommendations on employer hiring decisions. They find that

²We plan to estimate this model in a follow-up paper.

these features can steer buyers towards certain sellers and act as a substitute for costly screening efforts.

A key focus of the literature has been on the factors affecting hiring decisions in online labor markets. [Pallais \(2014\)](#) demonstrates the importance of information revelation on the platform O’Desk (now Upwork), showing that providing more detailed evaluations of workers’ job performance positively affects their employment outcomes. Providing minimal information about a contractor’s performance can dramatically increase their subsequent income. Studying O’Desk as well, [Stanton and Thomas \(2012\)](#) show that affiliation with an intermediary can substantially improve job-finding probabilities and wages for inexperienced contractors. [Ren et al. \(2023\)](#) find that employers on [Freelancer.com](#) prefer workers who share the same country of residence and have a preference for lower-cost workers over more experienced ones.

Our paper contributes to this literature by identifying a novel negative externality arising from competition in thick markets. We focus on the information provision trade-off faced by agents in thick markets, where many agents compete for the same matches, and highlight a fundamental tension between the potential efficiency gains from market thickness and the disincentives for agents to provide match-specific information. As markets thicken, there should exist more potential high-value matches to be found, as many argue that thickness is required for two-sided markets to function efficiently ([Roth, 1982](#); [Roth and Peranson, 1999](#); [Roth, 2008](#); [Ashlagi et al., 2020](#); [Arnosti et al., 2021](#)). However, we underline a key countervailing force: as markets thicken and competition increases, agents are disincentivized from providing match-specific information, threatening the potential efficiency gains of thick markets.

This tension between market thickness and information provision is particularly relevant in the context of online labor markets, where the cost of providing information (as an applicant) or screening applicants (as an employer) is a larger fraction of the overall value of a hire compared to traditional labor markets ([Krasnokutskaya et al., 2020](#); [Marinescu and Wolthoff, 2020](#)). [Krasnokutskaya et al. \(2020\)](#) assess buyers’ welfare gains from the globalization enabled by online labor markets, finding that access to international markets substantially improves buyers’ options. [Marinescu and Wolthoff \(2020\)](#) demonstrate the importance of job titles in the matching process, finding that higher wages attract more applicants within a job title but not within a detailed occupation, suggesting substantial applicant heterogeneity.

Our paper also relates to the literature on congestion as a friction in market design ([Roth, 1982](#); [Roth and Peranson, 1999](#); [Roth, 2008](#); [Ashlagi et al., 2020](#); [Arnosti et al., 2021](#)). We build upon these findings by focusing on the information provision trade-off

faced by agents in thick markets and highlighting the importance of finding informative signaling technology to mitigate this tension and improve matching outcomes. By combining insights from the literature on congestion as a friction in market design and empirical studies of online labor markets, our paper contributes to a better understanding of the challenges and opportunities in designing efficient online labor markets. By identifying the negative externality of competition on information provision and the resulting tension between market thickness and efficiency, we provide a foundation for future research on mitigating this tension and optimizing online labor market design.

Empirical Auctions. Second, we contribute to the literature on empirical auctions, which has seen significant developments in recent years, with a growing focus on scoring auctions and multidimensional bidding environments. Our work contributes to this literature by studying a scoring auction with multidimensional bidder actions and non-deterministic scores, extending the seminal work of [Che \(1993\)](#) and the subsequent analyses by [Asker and Cantillon \(2008\)](#).

[Che \(1993\)](#) introduces the concept of two-dimensional auctions, where firms bid on both price and quality, and bids are evaluated using a scoring rule designed by the buyer. He shows that if the buyer can commit to a scoring rule in his best interest, the resulting optimal scoring rule under-rewards quality relative to the buyer's utility function and implements the optimal outcome for the buyer under first- and second-score auctions. Che's work has been instrumental in understanding the design of scoring auctions and the trade-offs between price and quality in procurement settings. Building upon Che's framework, [Asker and Cantillon \(2008\)](#) provide a systematic analysis of equilibrium behavior in scoring auctions when suppliers' private information is multidimensional. They further explore the properties of scoring auctions, deriving the optimal buying mechanism when quality matters and demonstrating the advantages of scoring auctions over alternative procurement methods.

Our work extends these analyses by embedding a signaling model into a scoring auction framework, allowing us to study the interplay between multidimensional bidding and the provision of match-specific information by bidders. By incorporating non-deterministic scores, we capture the notion of uncertainty in the buyer's evaluation process and provide a more realistic representation of real-world procurement settings.

Our work also builds upon the seminal contribution of [Guerre et al. \(2000\)](#) in the field of structural estimation of auction models. They propose a general approach and a computationally convenient estimation procedure for the structural analysis of first-price sealed-bid auction data within the independent private value paradigm. Their work opened the way for nonparametric estimation of auction models, by showing that

the underlying distribution of bidders’ private values is identified from observed bids and the number of actual bidders without any parametric assumptions. By extending their nonparametric identification results to our multidimensional setting, we are able to recover the primitives of the environment and quantify the impact of information provision on bidding behavior and auction outcomes.

Finally, our work is closely related to the study by [Krasnokutskaya et al. \(2020\)](#) on the role of quality in internet service markets. They develop a methodology to recover primitives of the environment in the presence of unobserved seller heterogeneity in online procurement markets, while accommodating buyer-specific choice sets and high seller turnover. Although their focus is on assessing buyers’ welfare gains from globalization in online markets, their approach to handling unobserved heterogeneity and endogenous participation by sellers is relevant to our setting, where bidders’ quality and outside options play a crucial role in shaping equilibrium outcomes.

Information Economics. Third, our work makes a significant contribution to the field of information economics by demonstrating how large language models (LLMs) can be used to process text data, making it possible to estimate communication games empirically. [Spence \(1973\)](#) seminal paper laid the foundation for this literature by introducing a model in which more able individuals signal their ability to be paid more. Despite the richness of the theoretical literature on signaling, empirical studies in this area have been relatively scarce. One notable exception is [Tô \(2018\)](#), which provides evidence of the labor market consequences of parental leave-taking due to signaling and highlights the importance of asymmetric information in shaping parental leave choices.

While there are numerous theoretical models of communication games, the difficulty in quantifying communication has impeded the empirical application of these models. Our work shows that LLMs can be employed to transform text data into quantifiable measures of communication, enabling researchers to test the predictions of theoretical models of signaling and communication games in real-world settings and opening up new avenues for research in this area.

Outline. The rest of the paper proceeds as follows. Section 2 describes how our market of focus, [Freelancer.com](#), operates, and the data that we use. Section 3 presents our equilibrium model of information provision. Section 4 builds upon the prior two sections by presenting reduced-form evidence of the model implications in the data. Finally, Section 5 concludes.

2 Market Description and Data

This paper uses new data from `Freelancer.com`, a major online labor market. The first part of this section describes the market structure, including the platform’s players, the type of tasks outsourced, and the job matching process. The second part details the structure of the data we obtained through our collaboration with `Freelancer.com`.

2.1 Market Description: A Global Digital Marketplace

`Freelancer.com` is one of the world’s largest web-based job marketplace by number of users and posted jobs. Digital marketplaces for freelance labor, such as `Freelancer.com` or `Upwork`, match global workers with potential employers around the world to trade tasks that are delivered remotely, such as software development, sales and marketing support, and creative work - what is called “freelancing”. Since its creation in 2009, more than 71 million users from over 247 countries, regions and territories signed up for remote work on this globalized market. In fiscal year 2019 alone, over USD 123 million in gross value of payments were made through the platform. With an average payment per job of USD 190, this means that 650,000 job tasks were completed on the platform on average in 2019.

2.1.1 A North-South Trade in Tasks

Digital marketplaces globalize traditionally local labor markets, with platforms that enable employers primarily located in high-income countries to easily outsource tasks to contractors, primarily located in low-income countries. They also provide workers in the Global South with work opportunities that may not have been offered in their local labor markets. `Freelancer.com` explicitly offers low barriers to entry for workers (a main difference with competing platforms like `Upwork`).³ The outsourcing in tasks on this market is thus characterised by a North-South exchange, with most employers located in the Global North and most workers in the Global South.

Figure 1 compares the geographical distribution of workers and employers on the platform. The top panel displays the top 10 countries of employers. 50% of all jobs are posted from employers located in the United States (25.5%), Australia (11%), the United Kingdom (9.5%) and Canada (5%). While India also appears to be a large poster of jobs

³Upwork set up a screening process for prospective workers. This screening includes reviewing the freelancer’s profile, portfolio, and sometimes a verification interview. The platform’s claimed objective of screening prospective workers is to maintain high quality of services and to ensure that the supply of freelancers matches client demand in various job categories.

on the platform (8.5%), it is above all the primary country of origin of workers: 28% of all freelancers are from India. Pakistan (7%), the Philippines (5%), Indonesia (4%) and Bangladesh (3%) follow.

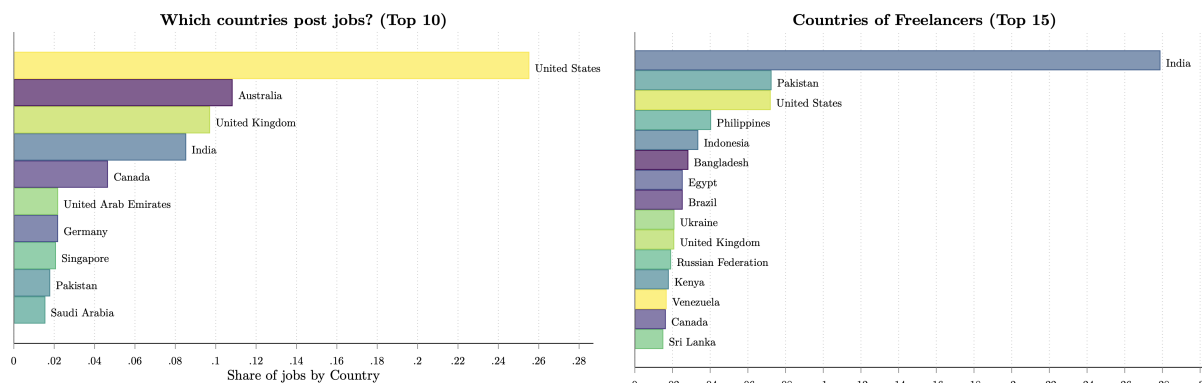


Figure 1: Who is on the platform?

Notes: These figures plot the listed country of workers (“freelancers”, on the right) and employers (individuals or firms posting jobs, on the left). Firms on the platform are largely from the Global North, while freelancers on the platform are mostly from the global south

2.1.2 Types of Jobs Posted

Online labor markets combine two increasingly important pillars of the modern economy. First, they are digital platform markets — marketplaces in which a third party facilitates matches between buyers and sellers. Second, they are largely centered around gig work — tasks that firms or individuals outsource to independent workers. These jobs can be done remotely and require little capital other than a computer. Workers on Freelancer.com are mostly hired to perform jobs in areas such as software development, writing, data entry and design (Figure 2). Each broad occupation can be further disaggregated into detailed occupations and skills. For a detailed list of all possible skills and job categories, see Appendix Table F1.

2.1.3 How do Employers and Workers Match for Jobs?

Figure 3 describes the two-sided market’s dynamics of the platform. The job matching process can be decomposed into five main steps.

In a first step, firms or individual employers (“clients”), looking to outsource a specific task, first post jobs (“projects”) on the Freelancer.com platform. In a second step, workers with matching skills to those required for the task place bids on the job project. These bids include two components: first, how much they are asking to be paid for the

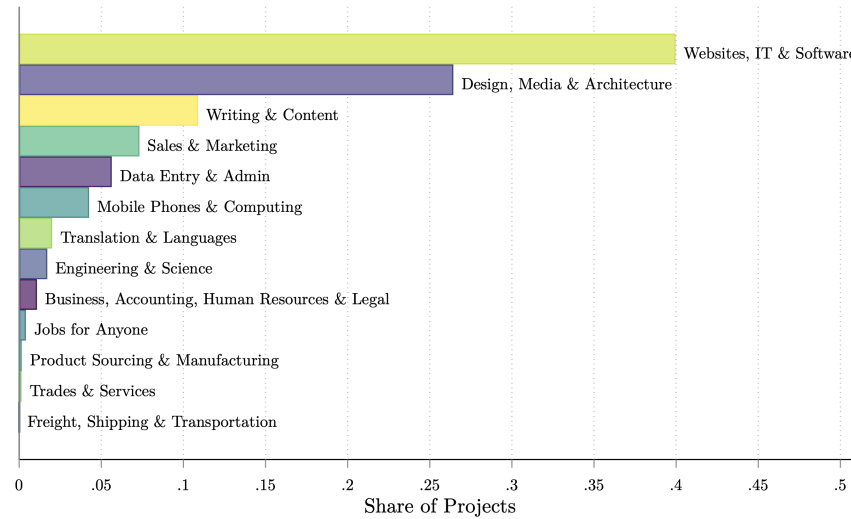


Figure 2: What Types of Jobs Are Posted on the Platform?

Notes: This figure plots the aggregate job categories listed on the platform. Coding, design, and writing skills are most in demand on the platform. For a detailed list of all possible skills and job categories, see Appendix Table F.1

task and second a written job proposal functioning as a cover letter. In a third step, given the current set of competitive bids and the characteristics of the workers competing for the tasks, the platform ranks bids for employers according to a recommendation algorithm.⁴ In a fourth step, employers then choose the worker with the most attractive bid elements to complete the task, and work commences. Employers can also choose the outside option of not picking anyone and leaving the job unfilled. In the fifth and last step, after completion of the job, both the worker and the employer can evaluate the quality of the job done by attributing a “star rating” to the other party. Payments are managed through the platform and broken down into “milestones” steps.

Several other characteristics of workers’ bids and employers and workers’ profiles are important to underline.

Searching for Jobs. Figure F.1 provides a screenshot of the platform’s appearance from the point of view of workers looking for jobs. Workers can search job postings by budget and budget type (either fixed-price projects or hourly projects), by project duration, by language or by job categories and skills. Within a given job category, jobs are automatically ranked by posting time, with the newest jobs ranked first. Workers can also sort jobs by budget or number of bids/entries. Each job posting comes with a description of the job tasks and timing, a list of skills necessary for the project, as well

⁴The algorithm behind the recommendation system is not public information and changed over time. Freelancer.com provided us with their recommendation algorithm for the purpose of this project.

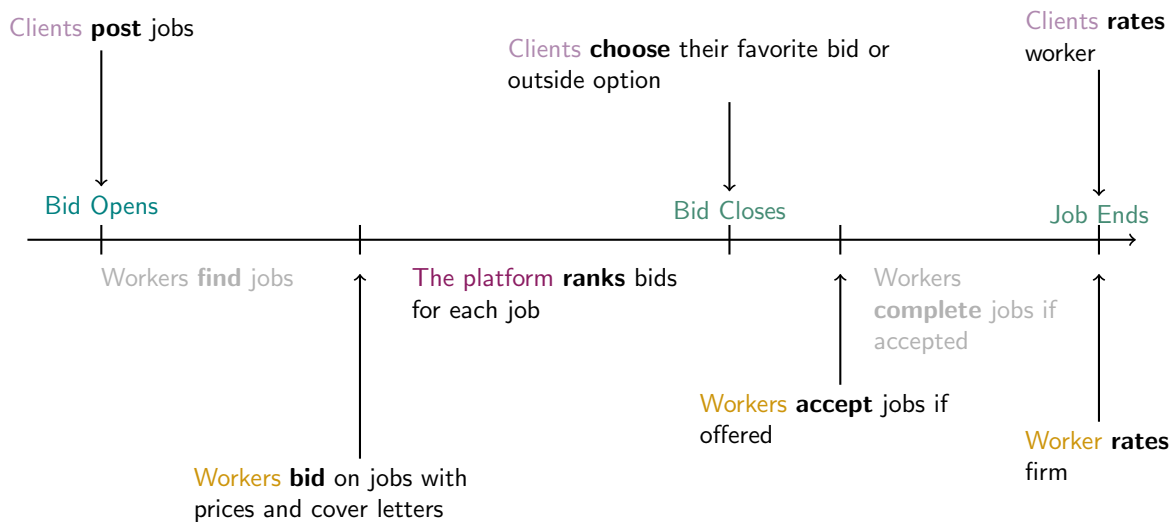


Figure 3: Matching Process

Notes: This figure describes the process during which employers and workers match for jobs on Freelancer.com. In a first step, firms or individual employers (“clients”), looking to outsource a specific task, first post jobs (“projects”) on the Freelancer.com platform. In a second step, workers with matching skills to those required for the task place bids on the job project. These bids include two components: first, how much they are asking to be paid for the task and second a written job proposal functioning as a cover letter. In a third step, given the current set of worker competing for the tasks, the platform ranks bids for employers according to a recommendation algorithm. In a fourth step, employers then choose the worker with the most attractive bid elements to complete the task, and work commences. Employers can also choose the outside option of not picking anyone and leaving the job unfilled. In the fifth and last step, after completion of the job, both the worker and the employer can evaluate the quality of the job done by attributing a “star rating” to the other party.

as the current number of bids for this job and the average bid amount from competitors.

Workers and Employers Information. Before selecting a worker for a job, employers can see the information on the freelancer’s bid card as well as the freelancer’s profile if they click on their name (see Figures F.3 and F.2).⁵ Workers can also see some limited information about the clients when viewing the project details (see Figure F.4).

Accepting a Job. Bids posted by workers during the auction process are not binding. Workers have 46 hours to accept or reject a project after being awarded a job from an employer. If no action is taken by the worker within 46 hours, the platform considers that this is an implicit reject. Rejections remain relatively rare.

Employers-Workers Communication Tools. Platform-guided recommendations and

⁵Employers looking for a worker to complete a project can either publicly post a project and open it for competitive bidding, or they can browse through freelancer profiles and directly send a request for hiring to a specific worker (see Figure F.2). However, the posting of a fixed-price project is the vastly predominant mode by which employers look for workers. The “Freelancers for Hire” section was not a prevalent feature of the platform in 2019-2020.

direct communication are key components of matches. There are no interviews by employers post-acceptation of a project. Before selection however, both employers and workers can communicate in a decentralized, private chat system hosted by the platform.

Upgrades and Memberships. Bidding for a job is free of charges, but there is a finite number of bids you are able to make as a worker. Membership plans allow to increase the number of monthly bids that workers can make (see Figure D.3 for a detailed description of available plans and pricing, and Figure F.5 for a screenshot of the current membership page). Freelancer.com also offers an upgrade system that allows both employers and workers to improve the visibility and appeal of their projects or proposals through paid features. These upgrades include options for highlighting projects, marking them as urgent, or featuring them on the website’s homepage, thereby increasing the likelihood of attracting higher-quality bids from freelancers. Additionally, freelancers can use upgrades to promote their bids, making them more noticeable to potential clients. Upgrades are tied to a project, not to an employer or a worker.

Payment System. Freelancer.com directly manages secured payment transactions between parties through their milestone payment system (see pictures of payments in Figure F.5). This system operates as an escrow mechanism and ensures mutual trust between clients and freelancers. This arrangement allows funds to be securely held until predefined project milestones are achieved, at which point payments can be released to workers.⁶ Such a system not only guarantees financial security for the service provider but also offers clients control over the payment process, contingent on their satisfaction with the work delivered.

2.1.4 The Current Recommender System

Unlike most in-person labor markets, online labor markets are literally designed, which provides a rare opportunity for realistic labor market design. The current recommender system is designed around a deterministic function, primarily focusing on the historical work performance of workers on the platform. The main component of workers’ history that the platform uses to rank freelancer bids is their star rating (see example in Appendix Figure F.3) and their listed job categories and skills expertise. The algorithm is insensitive to the two main components of a worker’s bid: the bid amount and the quality of their job application proposal.

Although this approach rewards freelancers with established track records, placing

⁶Payments are not conditional on achieving a required quality threshold.

them at the forefront of bid rankings, it also inherently makes it challenging for newcomers to gain visibility. The ranking algorithm rewards good bidders rather than bad bids, regardless of the merit or competitiveness of their proposals.

2.2 Data Summary

We have access to all data recording activity on the platform. The data we use can be classified into 8 categories.

The two main categories of data include job postings (title, description, timestamps, user ID (not identity) of poster, etc.) and job applications, i.e., bids (written bid proposals, dollar value of bid, time value of bid, timestamps measuring how long each worker spends writing their bid proposals, user ID (not identity) of bidder, etc.).

We also have information on job completion and payments (which bidder, if any, was selected to complete the project, ex-post ratings of the workers and clients, timestamps, tasks completion status, etc.), and on the work history of jobs completed on the platform.

Finally, we have data on workers (freelancers) gender and country, on observable worker profiles, and on user-interface data such as clicks data. This last category allows us to compute “consideration sets” for employers (which workers were considered) and workers (which jobs did they consider applying to).

Additionally we have access to information on past implementations of certain features of the platform design and other institutional information from `Freelancer.com` on how the platform is designed and run – for example, its bid proposal ranking algorithm, which is the main object of study in our proposed simulation.

Appendix E further details the main variables used in this paper.

3 Model of Information Provision in a Two-Sided Markets

In this section, we present our model of information provision in a two-sided matching market with transfers. This model provides a framework for thinking about how congestion can incentivize workers to send less informative signals of their quality to employers. Moreover, this structural model provides an empirical framework that can be readily applied in future work.

The main contribution of our model is to embed a career concerns model of signaling (Holmström, 1999) into a scoring procurement auction in which employers care both about match quality and price. The key difference with the original Holmström

(1999) framework, apart from the auction model, is that we model signaling effort as magnifying the contribution of match quality to the overall signal output relative to the random error. By contrast, the original Holmström (1999) model only incorporates an error term, ability (which becomes match quality in our modeling framework), and an effort term that is additive to ability in order to produce the observed signal. As a result, in our model, if all agents sending signals increased their effort, signals will become overall more informative about match quality. Conditional on match quality, the more effort a worker puts into their job application, the better the signal will look like from the employer’s point of view. In contrast, if congestion incentivizes workers to lower their signaling effort in equilibrium, then congestion will make it more difficult for employers to learn about match quality. With insufficiently high prior beliefs about match quality, congestion will thus lead employers to choose the outside option more frequently, harming the ex-post efficiency of the market.

An outline of the model is as follows. We build a static model of a cross-section of different job postings. A set of workers randomly arrives at each job, and each of them applies to the job in the form of a monetary bid and a signal of match quality. Employers form beliefs about the match quality of each applicant based on the signal, and then choose a single worker or the outside option to maximize their expected utility. Once this choice has been made, the employer pays the chosen worker, or pays no one if the outside option was chosen, and payoffs are realized.

Note that our model accounts for dynamics in a static way. First, we treat a worker’s history as an observable that is fixed at the time of a bid. At the time a competitive auction for a job starts, each competing worker’s history is fixed. Second, the marginal value for a worker of an additional employer’s review is small. Empirically, getting one more review is not critical to obtain a job so we assume it is not first order to help a worker’s bid (see Section 4).⁷ Finally, our model’s focus is on the signaling game and how global workers supply information in online marketplaces.

3.1 Environment

Workers. Each worker (freelancer) i has a vector of discrete observables $x_i \in X$, including country, verification status, and reputation on the platform. We omit time subscripts as

⁷The weakest part of this argument may be when considering the marginal value of going from zero to one review. However, note that nothing prevents new prospective workers from creating an employer’s account and giving themselves a first job, or having a friend do so, in case acquiring a first review seems valuable. We furthermore find in Section 4 that the conversion rate from bid to job is low for all workers, not only new workers. Finally, most of our empirical analysis focus on the bidding behavior of already established workers on the platform.

we model the matching process as a static auction game. Each job (project) j has a vector of discrete observables $\omega_j \in \Omega$, including country of the employer (client), budget, and skills required. Note that we use j to index both employers and jobs, since we consider them one in the same.

A set of workers, I_j , simultaneously and randomly arrives at job j , where I_j is a $|X|$ -dimensional vector whose entries are the count of each worker of type $x \in X$ matched to j . Formally, $I_j \in \mathbb{N}^{|X|}$, and $I_j \sim i.i.d. F_{X,\Omega}(\cdot, \omega_j)$, where $F_{X,\Omega} : (\mathbb{N}^{|X|}, \Omega) \rightarrow [0, 1]$ such that for all $\omega \in \Omega$, $\sum_{I \in \mathbb{N}^{|X|}} F_{X,\Omega}(I, \omega) = 1$. We assume that each worker matched to job j submit an application.

Each worker has a match quality with the job η_{ij} , which is unknown to them and drawn i.i.d. according to CDF $F_\eta(\cdot, x_i, \omega_j)$, which we assume to be $N(m_\eta(x_i, \omega_j), V_\eta(x_i, \omega_j))$. The idea behind this assumption is that workers do not know exactly how their work would be valued by the employer, but they do have a correct prior belief distribution based on both their and the job's observable characteristics. Workers' observables then dictate their mean match quality $m_\eta(x_i, \omega_j)$, or vertical ability.

Employers. Each employer j receives ex-post utility from choosing bidder i :

$$U_{ij} = \beta(\omega_j) \cdot \eta_{ij} - \alpha(\omega_j) \cdot b_{ij} + v_{ij}, \quad (1)$$

where b_{ij} is worker i 's bid on job j , i.e. the payment j commits to pay i if j chooses i , v_{ij} is an idiosyncratic taste shock, known to j but unknown to i , drawn i.i.d. according to CDF $F_v(\cdot)$, which we assume to be a Type-1 extreme value (T1EV) distribution, and both taste parameters $\beta(\cdot)$ and $\alpha(\cdot)$ are allowed to vary by employer observables ω_j . Should an employer j choose the outside option, indexed by $i = 0$, she will receive utility v_{0j} , which is also drawn i.i.d. from a T1EV distribution. Note that this normalizes the mean value of the outside option.

Employers are attention-constrained and are often unable to consider all applications they receive. Instead, they engage in simultaneous search where they draw an attention constraint $\phi_j \stackrel{i.i.d.}{\sim} F_\phi(\cdot, \omega_j)$, with $\phi_j \in \mathbb{N}$. Each employer j is only able to consider ϕ_j applications.

Ranking Algorithm. The platform orders applications for employers according to a ranking algorithm which dictates which subset of applications employers will consider.

Each worker i will have a rank score

$$r_i \equiv r(x_i) + \zeta_i \in \mathbb{R}$$

, where $r(\cdot)$ is a real-valued deterministic function, mostly determined by workers' reputation on the platform, and ζ_i is a random component with a small variance used for tie-breaking, drawn according to CDF $F_\zeta(\cdot)$.⁸ The platform measures worker's reputation based on a combination of past reviews on the platform, or worker's history, which we treat as an observable that is fixed at the time of a bid.

We assume workers know $r(x_i)$ but not ζ_i . Each application's realized rank $R_{ij} \in \mathbb{N}$ is then computed as an ascending count starting from 1 such that $R_{ij} \leq R_{i'j}$ if and only if $r_{ij} \geq r_{i'j}$. For example, if $R_{ij} = 4$, then i had the fourth highest rank score of all workers applying to job j .

Each employer j thus considers an application from worker i if and only if $R_{ij} \leq \phi_j$. Note that conditional on observing workers' observables x_i , R_{ij} does not provide any additional information about match quality η_{ij} to employer j . The employer thus only uses the ranking to form consideration sets and not to learn anything about η_{ij} .⁹ Also note that workers do not know whether their applications will be considered since they do not know their exact rank score, the rank scores of their competitors, or the attention constraint of the employer. Workers, do, however form beliefs over their chance of being considered, which is a crucial component of our model we discuss in more detail later in this section.

Worker's Signaling of Ability. When each worker i applies to project j , they choose a signaling effort $e_{ij} \in (0, \infty)$ at a convex cost $\psi(\cdot, x_i, \omega_j)$, where for all x and ω , $\psi(\cdot, x, \omega)$ is a twice differentiable, increasing, and convex real-valued function with $\psi(0, x, \omega) = 0$. The signal that each worker i sends to employer j will also come with random noise ϵ_{ij} unknown to the worker and drawn i.i.d. according to CDF $F_\epsilon(\cdot)$, which we assume to be $N(0, V_\epsilon)$. Signaling effort e_{ij} and signaling noise ϵ_{ij} come together to produce the signal

⁸In reality, the algorithm is deterministic, but we model it with randomness to capture two features of the data that we omit from the model. First, we are binning some continuous variables in X so that each x_i is a vector of discrete variables. We argue that the small differences we don't capture here by binning continuous variables are effectively random and do not contain relevant information. Second, there are some rarely used variables that go into the rank score that we do not include in our model such as measures of trust that are mostly used to punish left-tail bad actors on the platform, e.g. those who have signed up for multiple accounts with the same IP address.

⁹Note moreover that the ranking algorithm is not high-dimensional in nature, and is thus not summarizing high-dimensional information that would be hard for the employer to process. Instead, the ranking is based on workers' reputation, as measured by their history on the platform, or "star rating".

s_{ij} :

$$s_{ij} = e_{ij} \cdot \eta_{ij} + \epsilon_{ij} \quad (2)$$

We thus model signaling effort as magnifying the contribution of match quality to the overall signal output relative to the random error. The multiplicative formulation of the signal is critical here; it constitutes the main departure from the original [Holmström \(1999\)](#) framework. As a result of the multiplicative formulation, if congestion incentivizes workers to lower their signaling effort in equilibrium, then congestion will make it more difficult for employers to learn about match quality. Workers will thus choose e_{ij} in order to imperfectly choose s_{ij} , which will be what the employer uses to form a belief about the pay-off relevant match quality η_{ij} .¹⁰

3.2 The Worker and Employer's Problems

Worker's private cost. A worker i selected by an employer to complete a project j would incur a cost c_{ij} drawn i.i.d. according to CDF $F_c(\cdot, x_i, \omega_j)$, where $F_c(\cdot, x_i, \omega_j)$ is absolutely continuous with density $f_c(\cdot, x_i, \omega_j)$ and support $[\underline{c}_j, \bar{c}_j] \subset \mathbb{R}^+$. Note that we let the bounds of these support vary by j only insofar as they vary with ω_j . This cost c_{ij} is known to workers, unknown to employers, conditionally independent of match quality η_{ij} , and independent of ϵ_{ij} . Thus, c_{ij} is the only private information worker i has about their type that the employer does not know. The motivation behind this assumption is that workers indeed have an exact sense of what they would do to complete the posted job, but they do not know how this work will be received.

Worker's choice of bid and effort. After drawing c_{ij} , each worker i applying to job j chooses a bid $b_{ij} \in (0, \infty)$ along with their signaling effort e_{ij} . In reality, included in ω_j is both a minimum and maximum bid allowed, $\underline{b}_j$ and $\bar{b}_j$ respectively, so each $b_{ij} \in [\underline{b}_j, \bar{b}_j]$. We assume that the range of possible costs, $[\underline{c}_j, \bar{c}_j]$ is such that the bounds $\underline{b}_j$ and $\bar{b}_j$ do not bind in equilibrium. This simplifies our analysis and accords with our assumption that all workers who arrive at job j submit an application. Future work could consider binding bid limits and a selection correction for those who arrive but do not submit applications.

¹⁰In separate empirical applications of this modeling framework, we measure signaling effort by the amount of time freelancers spend interacting with a job post before submitting an application, and we capture signals by defining a measure of writing quality that incorporates how well the written portion of the application responds to the job description.

Thus, each freelancer will choose e_{ij} and b_{ij} to maximize their utility:

$$P(j \text{ chooses } i | e_{ij}, b_{ij}, x_i, \omega_j) \cdot (b_{ij} - c_{ij}) - \psi(e_{ij}, x_i, \omega_j) \quad (3)$$

where $P(j \text{ chooses } i | e_{ij}, b_{ij}, x_i, \omega_j)$ is the probability of being chosen given the chosen level of effort, bid amount and observable worker and project characteristics, $(b_{ij} - c_{ij})$ is worker i 's net payment from project j conditional on being chose, and $\psi(e_{ij}, x_i, \omega_j)$ is the cost of signaling effort function. Note that workers do not observe their competitors when they arrive at a job posting. They only observe their own cost c_{ij} , their own observables x_i , and the job's observables ω_j . Therefore, the probability function in their expected utility accounts for the workers' uncertainty over their competitors' observables, their competitors' unobservables, their own unobserved match qualities, the employer's attention constraints, and all possible draws of signaling error and employer taste shocks. This probability also depends on the equilibrium play of all other players, which will be explored in the next subsection.

The Employer's Problem. We note that employer j does not observe match quality η_{ij} and only observes worker i 's bid b_{ij} and their signal s_{ij} . Thus, employer j 's ex-ante utility from choosing worker i will be:

$$EU_{ij} = \beta(\omega_j) \cdot \mathbb{E}[\eta_{ij} | b_{ij}, s_{ij}, x_i, \omega_j] - \alpha(\omega_j) \cdot b_{ij} + v_{ij}, \quad (4)$$

Thus, the employer's problem is:

$$\max_{i \in \{i' : R_{i'} \leq \phi_j\} \cup \{0\}} \{ \beta(\omega_j) \cdot \mathbb{E}[\eta_{ij} | b_{ij}, s_{ij}, x_i, \omega_j] - \alpha(\omega_j) \cdot b_{ij} + v_{ij} \}, \quad (5)$$

where $\eta_{0j} = b_{0j} = 0$ to represent that the outside option simply has value v_{0j} . Note that employers do not directly value worker observables x and only use them to form beliefs about match quality. The idea here is that ultimately employers only care about match quality, price, and idiosyncratic taste shocks.¹¹

¹¹Note that employment relationships on the platform are almost exclusively one-off; long continuing relationships are rare. As the focus of this paper is on the signaling game, our data sample does not contain auctions in which workers previously hired by prospective employers were participating.

The Worker's Problem. Accordingly, each worker's problem is:

$$\max_{(b_{ij}, e_{ij}) \in [b_j, b_j] \times (0, \infty)} \left\{ \underbrace{P \left(i = \arg \max_{i' \in \{i'' : R_{i''} \leq \phi_j\} \cup \{0\}} \left\{ \beta(\omega_j) \cdot \mathbb{E}[\eta_{i'j} | b_{i'j}, y_{i'j}, x_i, \omega_j, R_{i'j} \leq \phi_j] - \alpha(\omega_j) \cdot b_{i'j} + v_{i'j} \right\} \right)}_{\text{Probability that } i \text{ is chosen conditional on } i \text{ being considered}} \left| b_{ij}, e_{ij}, x_i, \omega_j, R_{ij} \leq \phi_j \right)} \times \underbrace{P(R_{ij} \leq \phi_j | x_i, \omega_j)}_{\text{Probability that } i \text{ is considered}} \times \underbrace{(b_{ij} - c_{ij})}_{\text{Net payment conditional on } i \text{ being chosen}} - \underbrace{\psi(e_{ij}, x_i, \omega_j)}_{\text{Upfront signaling cost}} \right\}. \quad (6)$$

Note that the probability that i is considered does not depend on i 's actions b_{ij} or e_{ij} . Thus, there is nothing any worker can do to affect their chance of being considered. They can only affect their chance of being chosen conditional on being considered. This fact is the key place where congestion enters the model. As the number of workers applying to each job increases, assuming firm attention constraints are held fixed, then the chance that any one worker is considered by the employer decreases. We will see in the next subsection that, in equilibrium, this effect will decrease workers' willingness to supply information to the market, i.e. they will lower their signaling efforts as congestion increases.

3.3 Equilibrium

3.3.1 Information Sets and Equilibrium Functions

We focus on symmetric pure strategy Bayes' Nash Equilibria (BNE).¹²

Workers' information. Workers know their own cost type c_{ij} , their observables x_i and the project's observables ω_j . They also know the distributions of competitors $F_{X,\Omega}$, costs F_c , match qualities F_η , signaling noise F_ϵ , taste shocks F_v , the random components of ranks cores F_ζ , and attention constraints F_ϕ – all of which are conditionally independent given x_i and ω_j . Finally, they know the equilibrium strategy each worker of type (c, x) would play, and how the employer computes $\mathbb{E}[\eta_{ij} | b_{ij}, s_{ij}, x_i, \omega_j]$.

Importantly, workers do *not* know their project-specific match quality η_{ij} , so that a worker's ability is not part of their private information under our setting. The fact workers do not privately know their types differentiates our signaling framework from [Spence \(1973\)](#). As actions do not reveal types, Spence's idea of pooling versus separating equilibria does not apply. [Appendix A](#) further discusses the key distinctions between our signaling model and [Spence \(1973\)](#).

¹²Note that while the uniqueness of the equilibrium could matter for counterfactuals, the model does not require to assume that this pure strategy BNE is unique.

Employers' information. The employer knows their own observables ω_j , the realized set of applicants I_j , the observables x_i of each applicant i , their own taste shock v_{ij} for each applicant i and their outside option taste shock v_{0j} . They also know their own realized attention constraint ϕ_j , the set of applicants for whom $R_{ij} \leq \phi_j$, the distributions of match qualities F_η and signaling errors F_ϵ , and the equilibrium strategy each worker of type (c, x) would play. Finally, they observe each worker's bid and signal (b_{ij}, s_{ij}) , and they compute $\mathbb{E}[\eta_{ij}|b_{ij}, s_{ij}, x_i, \omega_j]$ using Bayes' rule.

Bids and Costs Bijection. As usual in the auction literature, we restrict attention to strictly increasing and differentiable Bayesian Nash strategies (Guerre et al., 2000), so that the equilibrium choice of b is strictly increasing in c for all x and ω . This implies that there is a bijection between equilibrium bids and costs, and that when employers observe bids, they are able to back out each worker's cost. Since they know each worker's equilibrium signaling effort choice as a function of c , x , and ω , they then know the signaling effort that each worker chose in equilibrium, even though this effort is not directly observed.

Equilibrium Functions. Define $b^*(c, x, w)$ and $e^*(c, x, w)$ as the equilibrium bidding and effort functions, where $b^{*-1}(b, x, w)$, the inverse of $b^*(c, x, w)$, exists for every (x, w) according to the assumption above. Thus, when employers observe b_{ij} in equilibrium, they infer that worker i chose signaling effort $e^*(b^{*-1}(b_{ij}, x, w), x, w)$. We sometimes abuse notation and write this composite function as $e^*(b_{ij}, x, w)$.

3.3.2 Employer and Worker's Equilibrium Problems: Matching

Expected Match Quality. We can now compute the expected match quality $\mathbb{E}[\eta_{ij}|b_{ij}, s_{ij}, x_i, \omega_j]$, which can be rewritten as $\mathbb{E}[\eta_{ij}|e^*(b_{ij}, x_i, \omega_j), s_{ij}, x_i, \omega_j]$. For readability, we will omit x_i , ω_j , and the subscripts i and j from the following calculations, and we note that all of these calculations are done within discrete cells of x_i and ω_j .

First, note that $s|e^*(b)$ is distributed normally since it is a linear combination of two independent normally distributed random variables η and ϵ . Also note that under our multiplicative formulation of a worker's overall signal to employers (Equation (2)), match quality η and signaling noise ϵ are independent of the equilibrium signaling effort $e^*(b)$.

Next, we show that $(s|e^*(b), \eta)$ is bivariate normal. For any $(p, q) \in \mathbb{R}^2$,

$$p \cdot s + q \cdot \eta = (p \cdot e^*(b) + q) \eta + p \cdot \epsilon$$

, which conditional on $e^*(b)$ is another linear combination of two independent normally distributed random variables. Thus, we can conclude $(s|e^*(b), \eta)$ is bivariate normal.

Using this fact, we can write

$$\mathbb{E}[\eta|e^*(b), s] = m_\eta + \frac{\mathbb{E}[\eta \cdot s|e^*(b)]}{\text{Var}(s|e^*(b))} (s - \mathbb{E}[s|e^*(b)])$$

Note that since η and ϵ are independent,

$$\begin{aligned} \mathbb{E}[\eta \cdot s|e^*(b)] &= e^*(b) \cdot (V_\eta + m_\eta^2) \\ \text{and } \text{Var}(s|e^*(b)) &= e^*(b)^2 \cdot V_\eta + V_\epsilon \end{aligned}$$

Moreover, since $\mathbb{E}[\epsilon] = 0$, we can write $\mathbb{E}[s|e^*(b)] = e^*(b) \cdot m_\eta$. Thus, putting back in x , ω , and the subscripts i and j , and letting

$$\begin{aligned} A(b_{ij}, x_i, \omega_j) &\equiv \frac{e^*(b_{ij}, x_i, \omega_j) \cdot (V_\eta(x_i, \omega_j) + m_\eta(x_i, \omega_j)^2)}{e^*(b_{ij}, x_i, \omega_j)^2 \cdot V_\eta(x_i, \omega_j) + V_\epsilon} \\ \text{and } B(b_{ij}, x_i, \omega_j) &\equiv (1 - A(b_{ij}, x_i, \omega_j)) \cdot e^*(b_{ij}, x_i, \omega_j) \cdot m_\eta(x_i, \omega_j), \end{aligned}$$

we can write:

$$\mathbb{E}[\eta_{ij}|b_{ij}, s_{ij}, x_i, \omega_j] = A(b_{ij}, x_i, \omega_j) \cdot s_{ij} + B(b_{ij}, x_i, \omega_j). \quad (7)$$

Note that A and B are functions of bids b_{ij} , observables x_i and ω_j , and parameters $m_\eta(x_i, \omega_j)$, $V_\eta(x_i, \omega_j)$, and V_ϵ .

Employer's Problem in Equilibrium. We can use these derived functions to write out the employer's problem more directly:

$$\max_{i \in \{i' : R_{i'j} \leq \phi_j\} \cup \{0\}} \{ \beta(\omega_j) \cdot [A(b_{ij}, x_i, \omega_j) \cdot s_{ij} + B(b_{ij}, x_i, \omega_j)] - \alpha(\omega_j) \cdot b_{ij} + v_{ij} \}. \quad (8)$$

We consequently have written out the employer's discrete choice problem as a function of data, parameters to be identified, and the taste shock v_{ij} .

Worker's Problem in Equilibrium. The integral formulation of the worker's problem in equation (6), together with the fact that the underlying distributions are non-parametric means that, consistent with the auction literature, the equilibrium strategies

do not have an analytic solutions. Note that the classical auction literature (Guerre et al., 2000) provides a differential equation formulation of bidders' winning probabilities in an auction; however, this differential equation cannot be solved analytically.

A key methodological contribution of our model is to provide a reduced-form approach to this problem. The auction literature typically does not incorporate observable heterogeneity in bidders (as players typically do not exactly know who their competitors will be). Our rich dataset and signaling auction model allow us to incorporate such heterogeneity. While the probability of winning and its associated differential equation formula do not take an analytical form, we provide a numerical simulation-based approach to overcome this estimation challenge. We detail this approach below.

In the Bayesian Nash equilibrium, and thus holding fixed the equilibrium strategies of all other players, we can denote the ex-ante reduced-form probability that employer j considers worker i as $P^C(x_i, \omega_j)$. Similarly, we can denote the ex-ante reduced-form equilibrium probability that employer j chooses (hires) worker i conditional on employer j considering worker i as a function of bid b and effort e as $P^H(b, e, x_i, \omega_j)$. Note that since $P^C(x_i, \omega_j)$ does not take as inputs any equilibrium objects, and since we assume that all workers apply to jobs at which they arrive, $P^C(x_i, \omega_j)$ is an exogenous object to the equilibrium choices of bids and signaling efforts.

We can then rewrite the worker's problem from Equation (6) in equilibrium with these reduced-form equilibrium functions, as:

$$\max_{(b_{ij}, e_{ij}) \in [\underline{b}_j, \bar{b}_j] \times (0, \infty)} \left\{ P^H(b_{ij}, e_{ij}) \cdot P^C(x_i, \omega_j) \cdot (b_{ij} - c_{ij}) - \psi(e_{ij}, x_i, \omega_j) \right\}. \quad (9)$$

Proposition. *At the symmetric pure strategy Bayes' Nash Equilibrium, formulation (9) has an equivalent solution to the original worker problem (6), and it produces equilibrium strategies $b^*(c, x, w)$ and $e^*(c, x, w)$ that together with all assumed model distributions produce probabilities consistent with P^H .*

The intuition behind this reduced-form equilibrium probabilities is as follow. Because we are in a BNE with no private information, workers do not know how many competitors they will have, what their observables will be, or what their unobservable types will be. They also ignore the employers idiosyncratic preferences, how employers will form consideration sets, how their effort will turn into a signal, and how their actions and those of their competitors will turn into signals. However, workers have correct beliefs about the ex-ante distributions of these objects. We can then take an integral

of the joint distribution over all these objects and get a sense of workers' chances of winning for different bid and effort pairs. This allows us to estimate these probabilities non-parametrically on data.

Furthermore, as long as all competing workers are playing the same symmetric equilibrium (which they are, by definition of a BNE), workers' beliefs, which can be encapsulated by their beliefs over $P^H(b, e, x_i, \omega_j)$, will manifest in the actual play of the game when the data is large. Therefore, by identifying the actual $P^H(b, e, x_i, \omega_j)$ being played, and assuming all auctions are ex-ante identical conditional on ω_j , we are actually identifying each player's beliefs. We can thus interpret their choices via the FOCs (10) and (11) of the worker's problem (which is a function of their beliefs) to back out their cost and writing cost function.

Section 3.4.2 further details the interpretation and identification arguments behind these reduced-form probabilities.

First-order conditions and Congestion. We can now write down the two following first order conditions of the newly defined worker problem:

$$c_{ij} = b_{ij}^* + \frac{P^H(b_{ij}^*, e_{ij}^*, x_i, \omega_j)}{\left(\frac{\partial P^H(b, e, x_i, \omega_j)}{\partial b} \right) \Big|_{b_{ij}^*, e_{ij}^*}}; \quad (10)$$

$$\frac{\partial \psi(e, x_i, \omega_j)}{\partial e} \Big|_{b_{ij}^*, e_{ij}^*} = \left[\left(\frac{\partial P^H(b, e, x_i, \omega_j)}{\partial e} \right) \left(\frac{\partial P^H(b, e, x_i, \omega_j)}{\partial b} \right)^{-1} \right] \Big|_{b_{ij}^*, e_{ij}^*} \cdot P^H(b_{ij}^*, e_{ij}^*, x_i, \omega_j) \cdot P^C(x_i, \omega_j). \quad (11)$$

The first equation is the typical procurement auction first-order condition that shows how workers mark down their bids from their costs. The second captures how workers trade off the benefit from increasing their signaling effort above what the employer expects with the cost of that effort increase.¹³

As P^C decreases for all workers, congestion increases, and at its limit, P^C approaches zero. As made clear from the second first-order condition (11), as P^C approaches zero, the right hand side of that equation approaches zero, meaning equilibrium choices of e must also approach zero, as per the assumptions on ψ .

¹³Note that, as in (Guerre et al., 2000), the model is solved by simulation and we cannot establish any supplemental formal result or proposition about the properties of the equilibrium.

The logic behind these first-order conditions follows that of an all-pay auction. Since workers must pay for their signaling effort regardless of whether they are chosen, if they have a lower chance of even being considered in the first place, then they will be less willing to pay signaling costs up front. As all workers choose lower levels of signaling effort, due to the multiplicative structure of signal production, differences in signals between workers will become less informative as the signaling error will dominate match quality in signal production. In other words, as all workers provide less information to the market, employers will rely more and more on prior beliefs about match quality $m_\eta(x_i, \omega_j)$ to dictate choices. This effect can lead to significant ex-post hiring mistakes, and with low enough prior means of match quality for those workers likely to be in employers' consideration sets, employers will opt for the outside option more frequently, leading to inefficiently low match rates in the market.

Signaling Equilibrium and Signaling Effort Choice. We end this subsection by elucidating the logic behind the signaling equilibrium and the choice of signaling effort in the spirit of [Holmström \(1999\)](#). In general, the worker has an incentive to “fool” the employer by increasing effort above the equilibrium level. This incentive comes from the fact that increasing effort probabilistically increases the signal output without changing what effort level the employer assumes the worker chose, which leads the employer to increase her expectation of the worker's match quality, thereby increasing the worker's chance of being chosen. The equilibrium effort level, as described in equation 11, will thus be set such that any deviation upward will not be worth trading off against the increasing marginal cost of signaling effort.

Despite this incentive to “fool” the employer into thinking match quality is higher than it is, in equilibrium the employer is not fooled and know exactly what the worker's effort level is. Accordingly, it may seem paradoxical that the worker would willingly expend costly effort on signaling. However, since in equilibrium the employer expects a certain effort level, the worker is also dissuaded from lowering her effort below this level, since doing so would, following the same logic as above, decrease their chance of being chosen.

In the words of [Holmström \(1999\)](#), “ [T]he [worker] cannot fool the [employer]. Yet, he is trapped in supplying the equilibrium level [of effort] that is expected of him, because, as in a rat race, a lower supply of [effort] will bias the evaluation procedure against him.”

3.4 Identification

We present in this subsection the key identification propositions and arguments that show our signaling auction model is indeed identified. We leave the estimation and empirical implementation of our model to future work.

For ease of readability, we omit observables x_i and ω_j as well as discussion of consideration sets from what follows. Before this omission, we discuss how one would incorporate these into identification arguments.

Observables. Since we are using discretely binned observables x and ω , all of what follows can be thought of as being done within bins of $X \times \Omega$. Of course, the arrival distribution $F_{X,\Omega}$ itself can be identified by simply observing the realized distribution of observable types x that arrive at each job of observable type ω . The attention constraint distribution F_ϕ can be identified simply by having data on consideration sets ϕ for each job. We propose future work measures ϕ with click data from the employers showing how far down the ordered list of applications employers search.¹⁴

Consideration sets. The reduced-form consideration set probability P^C can be identified by observing the joint distribution of realized ranks R_{ij} , attention constraints ϕ_j , worker observables x_i , and employer observables ω_j . Since the following analysis will be within bins of $X \times \Omega$, P^C will just be a fixed probability. As a result, we simply define $P^*(b, e, x, \omega) = P^H(b, e, x, \omega) \cdot P^C(x, \omega)$ as the unconditional reduced-form probability of winning, and just write $P^*(b, e)$ below. Finally, we note that we can identify F_ζ by observing the rank scores r and observables x , such that we can identify $r(x_i)$.

Identification Strategy. Specifically, we are interested in identifying the distribution of match quality m_η and V_η , the distribution of signaling error V_ϵ , the distribution of costs F_c , the marginal cost function of signaling effort $\psi'(\cdot)$, and employer preferences β and α . We break up identification into sequential parts. We first detail the identification of our signaling model (section 3.4.1), before discussing identification of our auction model (3.4.2). We end this identification section by describing how we recover employer preferences, which are defined through a standard discrete choice model (section 3.4.3).

Before detailing the identification arguments, we provide an overview of the identification sequence which will inform our estimation routine:

1. Identification starts with our signaling model. From the signaling equation (2)

¹⁴As mentioned in model section 3.1, the ranking algorithm on the platform is not summarizing high-dimensional information that would be hard for the employer to process. Employers use the ranking to form consideration sets and not to learn anything about η_{ij} .

$s_{ij} = e_{ij} \cdot \eta_{ij} + \epsilon_{ij}$ we can recover the distribution of η_{ij} and ϵ_{ij} parametrically.

2. From the auction model, we can recover costs c and ψ non-parametrically from the data.
- 2.5 Using data on submitted bids and effort measures, we then recover the equilibrium function $e(b)$.
3. Combining steps 2.5 and 1., we can now estimate the employer's discrete choice problem over ex-post bids (equation 8).
4. Using as inputs the distributions of η_{ij} and ϵ_{ij} , we can estimate the employer's discrete choice model and recover from the data β , α and the distribution of ν (non-parametrically).

Note that the sequentiality of the identification argument has to do with the independence of its different parts. Let's take the baseline formulation of the employer's discrete choice problem of selecting a worker over ex-post bids, $EU_{ij} = \beta(\omega_j) \cdot \mathbb{E}[\eta_{ij}|b_{ij}, s_{ij}, x_i, \omega_j] - \alpha(\omega_j) \cdot b_{ij} + \nu_{ij}$ (equation 4). ν and e are both assumed i.i.d and orthogonal to one another. Effort e and bids b are equilibrium objects. Conditional on having data about which worker is picked, the bid amounts b , effort e and the signal s , the model is identified as long as there is enough variations in e and b (i.e. no mass data). Without heteroskedasticity, all we need to be identified is the variance of e (an endogenous model variable) to not be 0. This means that this equation can be parametrically identified as long as we have an equilibrium in which workers are not choosing a degenerate e distribution. For this to be the case, we need to put an assumption on the primitives of the mode such that this condition holds. We assume here that the distribution of effort is not degenerate, which is both a standard assumption in the IO literature and something we directly observe in the data (see Section 4). Conditional on having data about which worker is picked, the bid amounts b , effort e and the signal s , the model is thus identified as long as there is enough variations in e and b (i.e. no mass data).¹⁵.

3.4.1 Signaling Model

First, we begin with the simplest part of the identification argument – the signaling portion of the model. We will observe the joint distribution of (s_{ij}, e_{ij}) , and recall that η_{ij} , e_{ij} , and ϵ_{ij} are all independent. Thus,

$$\mathbb{E}[s_{ij}] = \mathbb{E}[e_{ij}] \cdot \mathbb{E}[\eta_{ij}] + \mathbb{E}[\epsilon_{ij}] = \mathbb{E}[e_{ij}] \cdot m_\eta$$

¹⁵We measure s as “writing quality” in empirical applications in section 4

, and thus m_η is identified:

$$m_\eta = \frac{\mathbb{E}[s_{ij}]}{\mathbb{E}[e_{ij}]}.$$
 (12)

Next, note that

$$\text{Var}(s_{ij}) = \text{Var}(e_{ij} \cdot \eta_{ij}) + \text{Var}(\epsilon_{ij}) + 2\text{Cov}(e_{ij} \cdot \eta_{ij}, \epsilon_{ij}) = \mathbb{E}[e_{ij}^2] \cdot V_\eta + \text{Var}(e_{ij}) \cdot m_\eta^2 + V_\epsilon.$$

Substituting $\frac{\mathbb{E}[s_{ij}]}{\mathbb{E}[e_{ij}]}$ for m_η and rearranging, we can conclude that we identify the following linear combination of V_η and V_ϵ :

$$\mathbb{E}[e_{ij}^2]V_\eta + V_\epsilon = \text{Var}(s_{ij}) - \text{Var}(e_{ij}) \left(\frac{\mathbb{E}[s_{ij}]}{\mathbb{E}[e_{ij}]} \right)^2.$$
 (13)

Moreover, note that $\text{Var}(s_{ij}|e^2) = e^2 \cdot V_\eta + V_\epsilon$, so putting this together with the last equation, we can see that both V_ϵ and V_η are identified.

The intuition behind these arguments follow the usual intuition from identifying linear models. Since ϵ_{ij} is mean zero, η_{ij} , e_{ij} , and ϵ_{ij} are independent, and we observe the means of e_{ij} and s_{ij} , we identify the m_η just by taking means. Note that this makes m_η sensitive to the measurement of e_{ij} and s_{ij} . We avoid the importance of this sensitivity by letting β be a free parameter. In particular, the level of η only matters insofar as it factors into employer preferences, so if we arbitrarily doubled m_η , a correctly specified model would halve β .

3.4.2 Auction Model

With m_η , V_η , and V_ϵ identified, we move to the auction portion of the model to see how bidders' strategies can be used to identify the worker's costs and marginal signaling effort costs.

Ideally, we would follow a GPV-style (Guerre et al., 2000) argument applied to the workers' reduced-form first order conditions to recover F_c and $\psi'(\cdot)$. However, the first barrier is that unlike typical empirical auction environments, we do not have that the probability of winning is equal to the probability of having the lowest (or highest) type.

We fail to have this condition for two reasons. First, we have two dimensional types,

one of which is unknown to the workers. So while it is the case that the worker with the lowest cost will bid the highest, they will not necessarily win, since they may not have the highest match quality. Second, even if a worker had the lowest cost and the highest match quality, they still may not win, given they may get an unlucky draw of signaling error lowering their signal and thus lowering the employer's belief about their match quality.

As such, we cannot analytically compute the probability of being chosen. However, since we observe the joint distribution of e_{ij} , s_{ij} , b_{ij} , and whether or not a worker is chosen, we can identify $P^*(b, e)$ at the current equilibrium evaluated at the observed combinations of (b_{ij}, e_{ij}) . These observed combinations will be the equilibrium pairs of (b_{ij}^*, e_{ij}^*) . Note that this does not solve our problem yet, since we do not just need to know $P^*(b, e)$ evaluated at equilibrium pairs, but also its two partial derivatives evaluated at those same pairs as per the workers' reduced-form first-order conditions.

To identify these derivatives, we however need to identify $P^*(b, e)$ at all pairs of (b, e) in open sets around each observed (b_{ij}, e_{ij}) . Conveniently, our model allows us to do just that.

First, we note that $P(j \text{ chooses } i | b_{ij}, s_{ij}, e_{ij}) = P(j \text{ wins } i | b_{ij}, s_{ij})$. Since the employer only observes bids and signals, they only choose their favorite application based on those two observables and their idiosyncratic shock v_{ij} , which is assumed to be independent of everything else. Thus, once we observe b_{ij} and s_{ij} , we do not gain any more information about worker i 's likelihood of being chosen by employer j by observing e_{ij} . We abuse notation and write $P^*(b, y)$ to denote the reduced-form probability of winning at the current equilibrium as a function of the chosen bid b and the realized signal y . Thus, we can now write:

$$P^*(b, e) = \int_{-\infty}^{\infty} P^*(b, y) f_{y|e}(y) dy, \quad (14)$$

where $f_{y|e}$ is the PDF of the distribution of s_{ij} conditional on e_{ij} . Moreover, note that once we condition on effort e_{ij} , the signal s_{ij} is determined only by two i.i.d. random variables, η_{ij} and ϵ_{ij} , so we only need joint data on (s_{ij}, e_{ij}) to identify $f_{y|e}$.

Next we claim that we can identify $P^*(b, y)$. Given η and ϵ have full support on the real line, so will $s_{ij}|e_{ij}$ for each choice of e . Thus, we will observe the full support of the joint distribution of b_{ij} , s_{ij} , and whether i wins j . As such, we can identify $P^*(b, y)$. The logic here is that given how e_{ij} randomly and exogenously translates to s_{ij} , even though

we see a selected set of equilibrium pairs (b_{ij}, e_{ij}) , we see all pairs of (b_{ij}, s_{ij}) , and thus can use them to identify $P^*(b, y)$.

Putting the previous arguments together, we identify the right hand side of equation 14, and thus identify $P^*(b, e)$ over its entire domain, which assuming it is differentiable, allows us to identify its partial derivatives at all interior points.

The intuition behind this whole argument is as follows. Imagine a worker with type c deviates to an off-equilibrium bid, say b' , but maintains the same equilibrium effort choice for her type, say e . Let c' be the type of worker who chooses b' in equilibrium, and let e' be the equilibrium effort chosen by type c' workers. We thus observe the success of workers choosing (b', e') , yet we are trying to identify the counterfactual success of a worker who chooses (b', e) . Due to the randomness and full support of signals conditional on effort, we also observe the success of a worker whose ex-post bid equals (b', s) for all s . Thus, by re-weighting that success of ex-post choices (b', s) according to the resulting signal distribution conditional on effort choice e , rather than e' , we can recover the desired quantity.

Similarly, imagine a worker with type c maintaining b but deviating to some effort choice e' . We know the success of ex-post pairs (b, s) for all s , and so by the same re-weighting argument, we can recover the success of a counterfactual ex-ante choice (b, e') .

We are now ready to turn back to the workers' reduced-form first-order conditions. Using the conjectured monotonicity of the equilibrium bidding strategy $b^*(c_{ij})$, assuming $P^*(b, e)$ is differentiable in both arguments, and assuming $\frac{P^*(b_{ij}^*, e_{ij}^*)}{\left(\frac{\partial P^*(b, e)}{\partial b}\right) \Big|_{b_{ij}^*, e_{ij}^*}}$ will be monotonic in b_{ij} , we can follow GPV and write:

$$c_{ij} = b_{ij} + \frac{P^*(b_{ij}, e_{ij})}{\left(\frac{\partial P^*(b, e)}{\partial b}\right) \Big|_{b_{ij}, e_{ij}}}. \quad (15)$$

Therefore, for each observed (b_{ij}, e_{ij}) , we can use equation (15) to back out the cost type associated with each pair. This will allow us to non-parametrically identify F_c .

Similarly, we can write:

$$\psi'(e_{ij}) = P^*(b_{ij}, e_{ij}) \cdot \left[\left(\frac{\partial P^*(b, e)}{\partial e} \right) \left(\frac{\partial P^*(b, e)}{\partial b} \right)^{-1} \right] \Big|_{b_{ij}, e_{ij}}. \quad (16)$$

Therefore, for each observed (b_{ij}, e_{ij}) , we can use equation 16 to back out the marginal cost of signaling effort associated with each effort choice. Thus, we identify a differential equation in $\psi'(\cdot)$. Note that we cannot use these same arguments to identify the level of $\psi(\cdot)$. Yet, since we assume $\psi(0) = 0$, we have a boundary condition to solve the differential equation identified above and recover $\psi(\cdot)$.

The intuition for how we use these first-order conditions to identify F_c and $\psi'(\cdot)$ is straightforward. The level of bid each worker chooses relative to how the probability of winning responds to bidding identifies the underlying cost relative to which workers are bidding. And the level of effort each worker chooses relative to how the probability of winning responds to effort identifies how marginally costly it is to raise signaling effort.

3.4.3 The Employer Discrete Choice Problem

At this point, all we have left to identify is β and α . Recall that the employer's problem, defined in problem 8, is a typical discrete choice problem over ex-post bids (b_{ij}, s_{ij}) where the objective function is linear in s_{ij} , β , and α and nonlinear in b_{ij} , m_η , V_η , and V_e .

The relevant moments we can use here are how often the employer chooses the outside option, and which worker they choose when they do choose an inside option. The former allows us to identify the levels of α and β relative to the normalized distribution of outside option values, F_v . The latter allows us to identify the difference between α and β , as per the usual discrete choice identification arguments.

Therefore, we have identified all the desired parameters, and are in fact over identified in that we can use the various levels of $\text{Var}(s_{ij}|e)$ for different values of e . We hope these arguments can also provide the framework for an estimation routine that follows the same split of the model into the signaling, auction, and discrete choice components.

4 Reduced-Form Evidence : Information Provision Trade-offs in Thick Digital Marketplaces

This section builds on our model of information provision in a two-sided matching market and provides empirical evidence that labor congestion in task-based digital marketplaces can result in workers sending less informative signals of their fit to employers.

We provide suggestive evidence of the tension between market thickness and information provision in online labor markets. Specifically, we study how agents navigate the trade-off between providing match-specific information and the probability of forming

a match in thick markets, where many agents compete for the same jobs. Our findings suggest a novel negative externality of competition: as the number of bidders for a job increases, workers invest less effort in providing match-specific information to potential employers, resulting in high information frictions about match quality. This reduced effort in signaling fit highlights the tension present in thick markets. While increased market thickness can potentially lead to more high-value matches, it may also provide low incentives to workers from providing the necessary information for those matches to form.

For the remaining of this section, it is important to distinguish between the notions of competition and congestion in our context. Competition is measured through the number of bidders by auction, and the reduced provision of information on the platform happens from competition alone. Congestion is the fact that the employer cannot look at all the applications (and this is magnified by large volumes of bids per job postings).

Sample Selection. The evidence presented in this section relates to the time period spanning January 1st 2019 to February 29th 2020, for projects listed under the “PHP”, “Python” and “Data Entry” categories.¹⁶ We restrict to projects posted by U.S.-based employers, in English and offering payments in USD.¹⁷ This includes 704,310 bids for 23, 896 projects and 80,787 unique workers (freelancers). The largest category of job postings is “PHP” with 581,796 bids for 18,401 projects. Table 1 summarizes the sample composition. All posted projects are only awarded to one worker at most in the data.

Table 1: Sample Size: PHP, Python and Data Entry Jobs (2019)

Jobs Category	Number of Projects	Number of Workers	Number of Employers
Data entry	3,633	39,050	2,522
PHP	18,401	46,040	9,491
Python	2,804	9,764	1,479

4.1 Measuring Signaling Effort and Job Applications’ Content Fit

To quantify the effort a worker invests in tailoring their job application to a specific job post, we construct two sets of measures. This section describes how we empirically

¹⁶PHP is a server-side scripting language, necessary for creating dynamic and interactive websites or applications.

¹⁷We do not place any restrictions on the workers’ identity and countries of origin. We select employers within the same country (the U.S.) to ensure their preferences are comparable. Note that given the ratio of freelancers to employers, employers are generally not considered to be competing for workers.

measure workers' effort in tailoring their job applications (signaling effort) and writing quality, or "content fit", to quantify the effect of these variables in the matching process.

4.1.1 Measuring Signaling Effort

We empirically measure workers' effort in tailoring their job applications (signaling effort) for a *specific* job post by the time spent between a worker first clicking on a job post and submitting their application.

This "bidding time" measure captures not only the time spent writing the application but also the time spent reading and understanding the job requirements, which is an essential part of providing match-specific information. By considering the entire duration from the initial click to the final submission, we establish an upper bound on the effort exerted by the worker, as it is reasonable to assume that no one can think about what to write before actually viewing the job post.

Our primary goal with this measure is to capture the level of engagement workers have with each specific job post. Anecdotally, employers often complain that workers do not seem to read their job posts thoroughly and that applications can be overly generic, failing to address the specific requirements outlined in the posting. By quantifying the time workers spend on each application, we aim to assess the cost associated with providing match-specific information. This measure is thus distinct from a standard "writing quality" measure. Under this measurement criterion, a carefully written but generic application that has been copy-pasted by a worker through several auctions should be classified as "low effort".

Figure 9 presents the distribution of worker effort, as measured by the time spent on each application. We validate our effort measure in Table 17 and the associated binscatter figure 10, in which we compare the quality of bids submitted within a few seconds to those submitted after 2-3 minutes. This regression framework suggests that signaling effort meaningful captures a worker's engagement with the job posting.

4.1.2 Measuring Content Fit

We build a novel measure of "content fit" that reflects the extent to which the written component of the workers' bid aligns with the task description. This measure aims to capture the degree of customization in the job-specific application and to assess how well each application demonstrates the worker's *engagement* with the specific content of the task-based job post. This "content fit" measure corresponds to the match-specific "signal quality" s in our theoretical model.

Traditional natural language processing (NLP) approaches, such as n-grams or supervised machine learning models, are not well-suited for this task as they lack the ability to capture semantic meaning and do not offer targeted measure of “text fit”.¹⁸

To overcome these limitations, we employ a large language model (LLM) that takes both the job post and the application as inputs. By doing so, we effectively ask the LLM to evaluate how well the application matches the posting. However, we do not rely on the LLM to directly output a numerical score, as LLMs generally treat numbers as words, leading to inconsistencies. Instead, we ask the LLM a yes/no question about the targeted fit of the job application, and extract the underlying posterior probability of a “yes” response, which serves as our raw content fit score.

As this raw score falls between 0 and 1, we transform it into a measure that spans from negative to positive infinity. To achieve this, we apply a logit transformation to the raw LLM scores, resulting in our final content fit measure. Figures 4 show the distribution of the raw LLM-generated scores, while Figure 6 show the distribution of the logit-transformed scores in our sample.

We validate our content fit measure by comparing examples of low and high scores within and across job posts and demonstrate its predictive power for employer choices using a logit model. Appendix C provides more information on our LLM measure of “content fit” (signal quality), including details on the prompt used for measuring content fit (section C.2). Appendix Section C.3 provides an illustrative example of the LLM scoring measure for one job posting and associated applications.

¹⁸It is important to note that our content fit measure is not merely an evaluation of writing quality but rather a more comprehensive assessment of how well the application aligns with the job post.

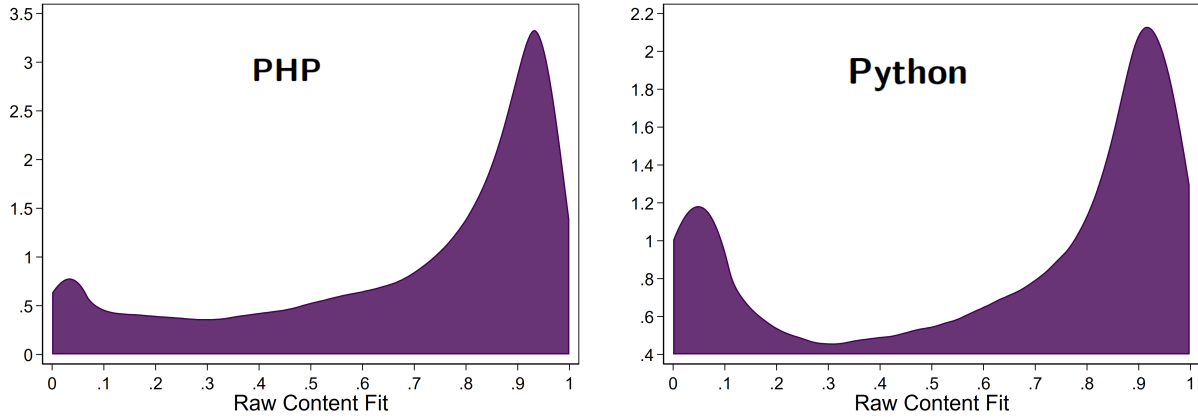


Figure 4: Raw “Content Fit” Distributions from the LLM

Notes: These figures plot the distribution of the raw “content fit” (signal quality) measure from the LLM. These raw scores fall between 0 and 1 (pre-logit transformation). Appendix Section C.3 provides an illustrative example of the LLM scoring measure for one job posting and associated applications. More complex, skill-intensive tasks (Python) are more likely to get a “no” from the LLM than less skill-intensive tasks (PHP).

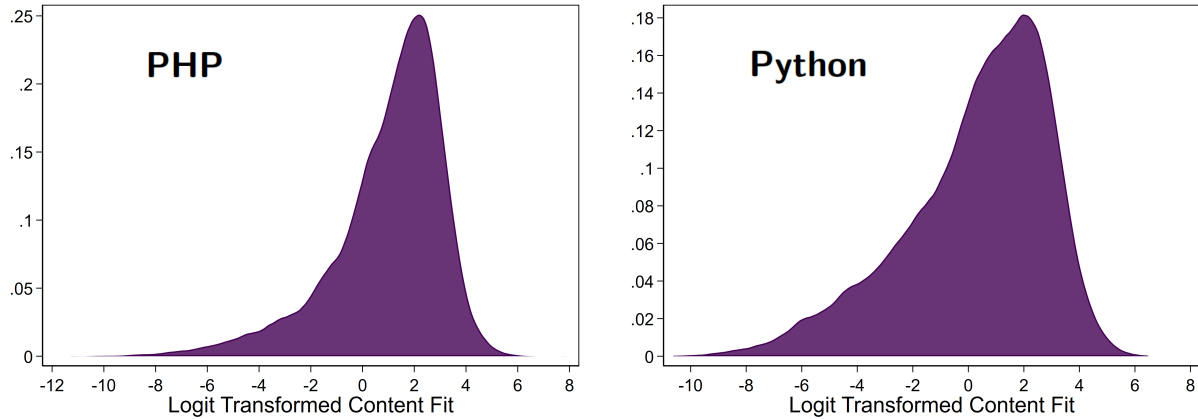


Figure 5: Logit-Transformed “Content Fit” Distributions

Notes: These figures plot the distribution of the “content fit” (signal quality) measure from the LLM, post logit transformation. These scores from negative to positive infinity. More complex, skill-intensive tasks (Python) are more likely to get a “no” from the LLM than less skill-intensive tasks (PHP), resulting in a fatter left tail after normalization. Appendix C provides more information on our LLM measure of “content fit” (signal quality), including details on the prompt used for measuring content fit (section C.2) and validation measures.

Our measures of effort (bidding time in minutes) and content fit (signal quality) are strongly positively correlated (Figure 6). Plotting the worker-project specific content fit scores against workers’ effort in the cross-section, we find that higher level of workers’

effort on an application are associated with higher signal quality, as measured by the content engagement of the job application letter to the specific job posting.

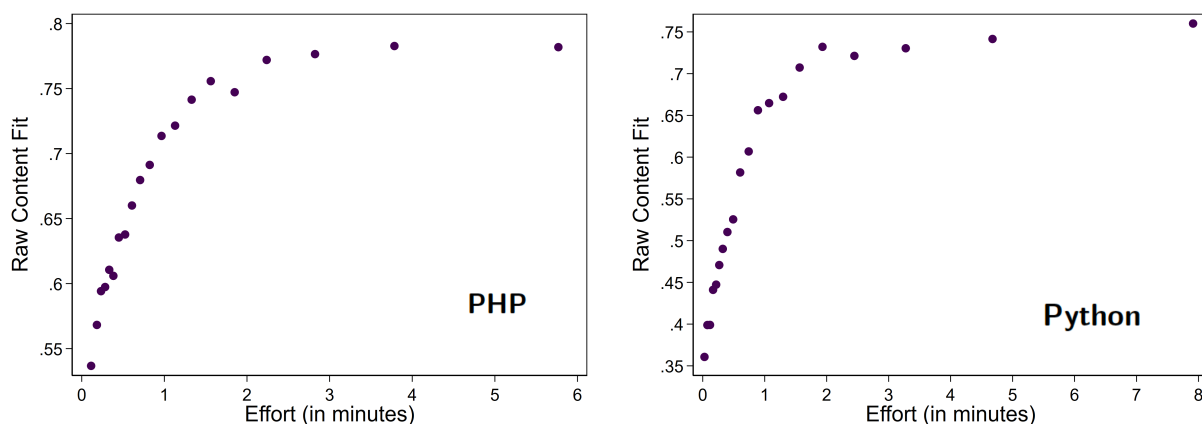


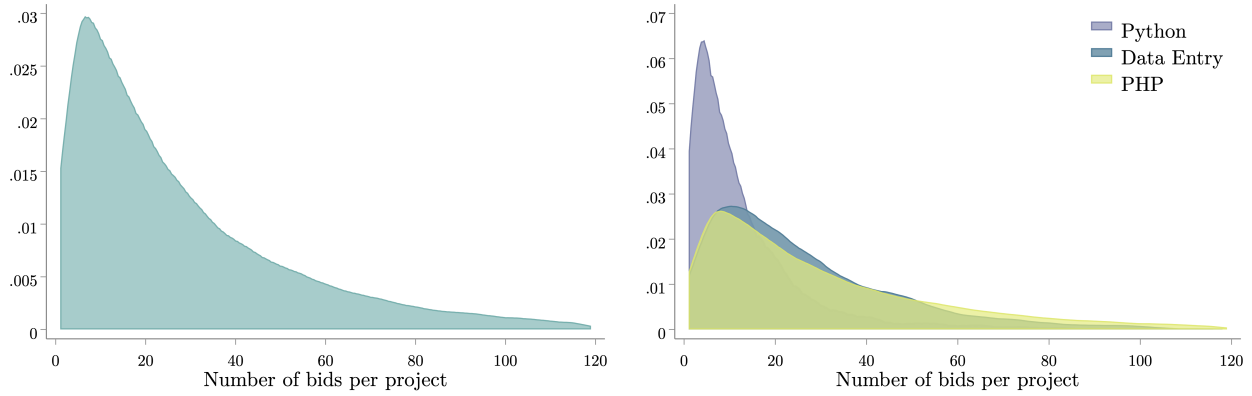
Figure 6: Raw Content Fit vs. Signaling Effort

Notes: These figures plot the distribution of the “content fit” (signal quality) measure from the LLM, post logit transformation. These scores from negative to positive infinity. Appendix C provides more information on our LLM measure of “content fit” (signal quality), including details on the prompt used for measuring content fit (section C.2) and validation measures.

4.2 Market Thickness and Competition

The Freelancer.com platform is characterized by substantial market thickness. Every job posted on the platform by prospective employers receives on average 31.6 bid applications (Figure 7) and the majority of these bids arrive within minutes after postings (Figure 8). The mean bid is submitted 4.3 hours after the job was posted, and the median bid is submitted within the first 6 minutes. By the end of the first day, 96% of projects posted on the platform have received the totality of their bids. Appendix Figure B.1 further splits fast bids from established freelancers with a good rating history (above 4-stars rating on a 5 stars scale) into 3 categories: bids submitted under 1 minute, under 5 minutes and under 10 minutes. It shows that bids from reputed workers tend to be written quickly.

This high volume of bids within a short time frame indicates intense competition among workers for the same job postings, a characteristic of thick markets.



Job Category	Mean	p50	sd	min	p25	p75	max	N
Data entry	27.54	21.00	24.95	1.00	11.00	36.00	313.00	3,633
PHP	32.46	23.00	30.94	1.00	11.00	44.00	370.00	18,401
Python	14.21	9.00	18.66	1.00	4.00	17.00	318.00	2,804
All Categories	29.68	20.00	29.55	1.00	9.00	40.00	370.00	24,838

Figure 7: Number of Bids by Job Posting

Notes: This figure plots the distribution of the number of bids received for each job (“project”) posted on the platform under the PHP job category. Bids are posted by prospective workers and are composed of a monetary part (the bid amount offered to perform the outsourced task) and a written job application. Every job posted on the platform by prospective employers receives on average 31.6 bid applications (median 22), with a standard deviation of 30.48

We find evidence that workers pay for membership plans that override the free-plan’s bid threshold and allow them to bid a lot (see Appendix F.5 for a detail of the plans). The distribution of bids made by workers every month evidence that the top percentiles of workers, as measured by the number of monthly bids submitted, bids above the 50 bids per month limit (Appendix Figures D.2 and D.3.)

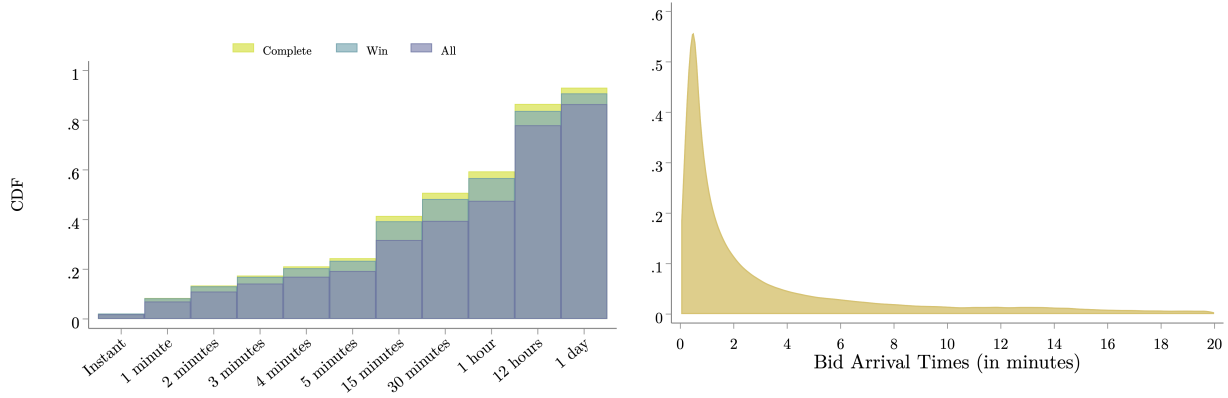


Figure 8: Large Volume of Bids Arrive Soon After Posting

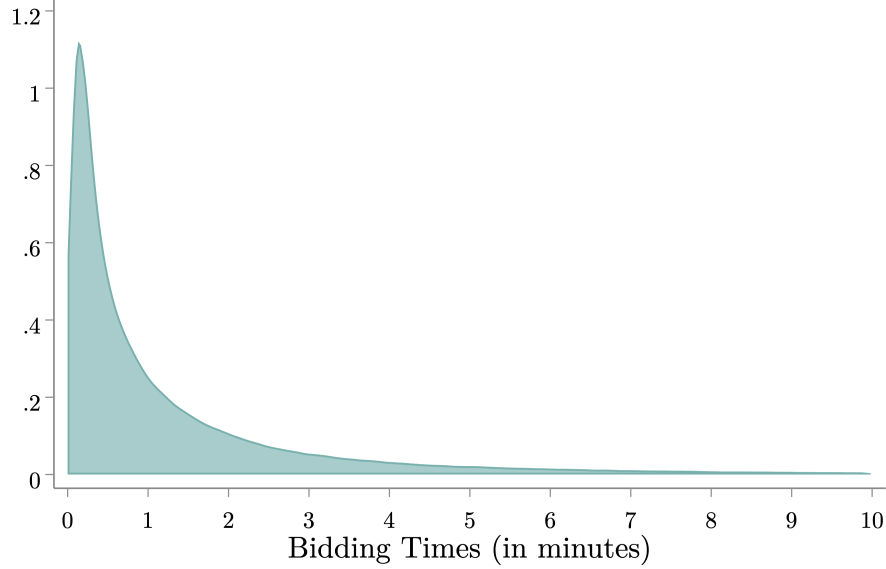
Notes: These figures plot the distribution of bids arrival time, measured as the time between a job ad was posted by a prospective employer and the time a worker’s bid was submitted. The right figure plots the entire distribution of bid arrival times, underlining that majority of bids are submitted under 10 minutes. The mean bid is submitted 4.3 hours after the job was posted, and the median bid is submitted within the first 6 minutes. After 2 hours, 75% of bids are received. The left figure further breaks down the bid submission times into discrete bin, and displays the CDF of bid arrival times by project completion status. The purple bins are for all projects listed on the platform. The blue bins are for job postings that were awarded to a freelancer (i.e. a winner as chosen among all bids). The yellow bins are for projects that have been completed (i.e. the worker successfully finished the outsourced tasks). By the end of the first day, almost all projects posted on the platform have received the totality of their bids. Projects that are awarded to a worker and completed projects tend to display more bids arriving after 15-30mn.

4.3 Competition and Information Provision

A high level of market competition (i.e. clients receiving many fast bids per job posting) goes in hands with low effort put in by workers into providing match-specific information. The average worker bids extremely fast: they only take 2 minutes to write and submit their bid proposal (with a standard deviation 4.56). Figure 9 displays the distribution of bids timing, measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid.¹⁹ Bids are composed of both a monetary amount and a written job application signaling job-specific fit. Our measure of bid timing thus encompasses both the time the worker spent reading the job

¹⁹Note that employers cannot observe this bid timing measure (“effort” in our setting). Employers can only observe bids arrival time (i.e. the time between the posting of their job listing on the platform and the time the bid was submitted). They only observe this arrival time if they are online at the time the bid was posted, or if they activated notifications for bids. A worker can submit a bid five hours after the auction starts but still write their bid under 2 minutes (which is our bid timing measure of “effort”). The only proxy the employer has to capture signaling effort is the writing proposal of workers, provided that longer and more customized job applications probably takes more time to write.

posting and the job proposal writing time. In Figure 9, 50% of all workers submit their job application within one minute of opening a job posting. The bottom 25th percentile submits bids within 0.22 minute and the top 5% within 8 minutes.



Job Category	Mean	p50	sd	min	p25	p75	max	N
Data entry	2.59	1.00	5.09	0.00	0.33	2.58	57.43	91495.00
PHP	1.98	0.60	4.67	0.00	0.22	1.72	57.52	576183.00
Python	2.34	0.77	5.08	0.00	0.25	2.13	57.20	29591.00
All Categories	2.08	0.65	4.75	0.00	0.23	1.83	57.52	697269.00

Figure 9: Workers Submit Job Applications Fast

Notes: This figure plots the distribution of bidding time, in minutes, for workers present on the platform in 2019. Bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid (composed of a monetary amount and a written job application signaling quality and fit). Bidding time thus encompasses both reading time for the job posting and job proposal writing time. On average, workers spend 1.96 minutes to submit their bid from the moment they click on a job posting (standard deviation 4.56). 50% of workers submit bids within 0.6 minute. The bottom 25th percentile submits bids within 0.22 minute and 95% of workers bid within 8 minutes.

Table 2 shows a strong positive relationship between the length of a job description (as written by prospective employers) and the time workers spend on their applications, validating that our measure of signaling effort meaningfully captures workers' engagement with the job posting. We regress our bid timing measure of effort on the text length of the job posting (in natural logarithm), controlling for the job category (PHP, Python or Data Entry) and whether the employer is new on the platform. The regression

specification is described in Equation 17 and includes a worker fixed effect γ_i , as well as fixed effects for the country of the bidders $\gamma_{country[i]}$ and the budget range of the posted job $\gamma_{budget[j]}$ (4 budget groups, in USD: “10-30”, “30-250”, “250-750” and “750+”). Figure 10 provides a binscatter visualisation of this regression. We find that despite the fact most workers spend little time writing and submitting their job applications, there are some substantial differences between writing a job proposal for a few seconds, 1 minute and 3 minutes.

These findings suggest that a high degree of competition in thick markets may act as a negative externality, providing low incentives to workers for investing in providing detailed, job-specific information.

$$\begin{aligned} \text{BidTiming}_i = & \beta_0 + \beta_1 \log(\text{JobPostLength}_j) + \sum_j \beta_{2j} \text{JobCategory}_j \\ & + \beta_3 \text{NewEmployer}_j + \gamma_{country[i]} + \gamma_{budget[j]} + \gamma_i + \epsilon_{ij} \end{aligned} \quad (17)$$

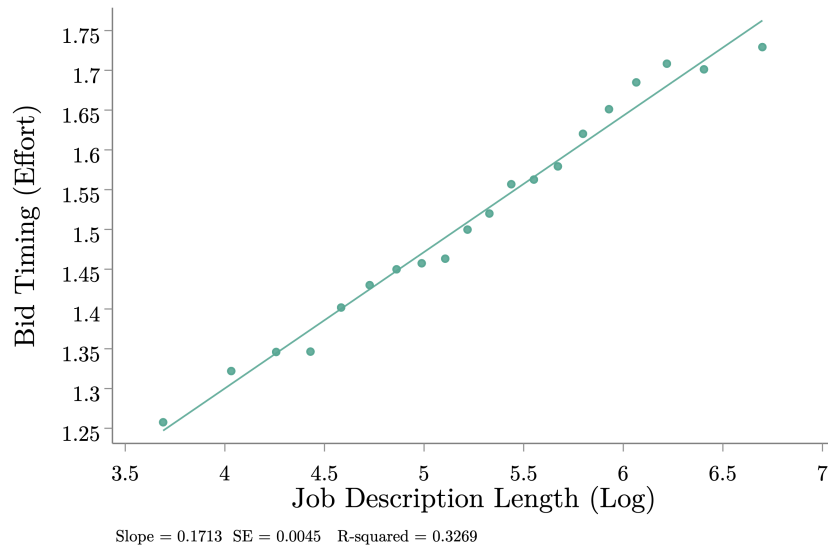


Figure 10: Job Posting Length and Bid Submission Time (Effort) are Positively Correlated

Notes: This binscatter figure displays the relationship between the time workers spend to bid for a job (what we call “effort”) in our model and the length of the job description ad, as written by the prospective employer. The length of a job posting description is strongly positively correlated with longer bid writing time from the workers side. It validates that the bidding time measure is a good predictor of writing time efforts. The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. Even though most workers submit their bids under 2 minutes, time spent writing is tightly correlated with an employer project description length (measured in number of characters).

Table 2: Job Posting Descriptions Length and Workers' Bid Submission Time (Effort) are Positively Correlated

	Dependent Variable: Bid Timing (Effort)			
	(1)	(2)	(3)	(4)
Job Description Length (Log)	.1856*** [.00509]	.1869*** [.00441]	.187*** [.00441]	.1713*** [.00454]
1{Employer is new on platform}			-.00326 [.00892]	-.03327*** [.00933]
Job Category				
PHP			.01767 [.01838]	-.00107 [.01847]
Python			.04655 [.0298]	.02386 [.03016]
Worker FE		✓	✓	✓
Worker's country FE				✓
Project's budget group				✓
Constant	.5802*** [.02644]	.5581*** [.02303]	.5418*** [.02892]	.6063*** [.02955]
F-Stat	1326	1795	449	361.9
Dep. Var.: Mean	1.557	1.542	1.542	1.498
Dep. Var.: Std. Dev.	2.536	2.523	2.523	2.458
N Obs.	3.7e+05	3.6e+05	3.6e+05	3.3e+05

Notes: * ($p < 0.1$), ** ($p < 0.05$), *** ($p < 0.01$). Standard errors are clustered at the FE level. This regression table displays the relationship between the time workers spend to bid for a job (what we call “effort”) in our model and the length of the job description ad, as written by the prospective employer. The length of a job posting description is strongly positively correlated with longer bid writing time from the workers side. It validates that the bidding time measure is a good predictor of writing time efforts. The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. Even though most workers submit their bids under 2 minutes, time spent writing is tightly correlated with an employer project description length (measured in number of characters).

4.4 Information Provision and Market Congestion

A potential downside of market thickness is that searching and being selected for jobs on the platform can be difficult, resulting in low matching rates. In our data, around 90% of all freelancers who joined between 2016 and 2018 never find a job, despite

submitting, on average, over 5 bids per month for 4.5 months before exiting the market. In our sample, 60% of projects posted in 2019 did not get awarded to a worker, despite receiving on average 30 bids. While new prospective workers on the platform get less than 0.01 job per submitted bid, even experienced workers having already completed some jobs and having more than a 4 stars rating (on a scale of 5) get on average 0.02 job per bid (Figure B.3). This situation is not unique to Freelancer.com; researchers estimate that out of the over 163 million workers who signed up for online freelancing, only 14 million of them have ever received work (Kässi et al., 2021).

On the employer’s side, high volumes of low-effort bids can make it challenging and time-consuming to identify the most suitable candidate for the job, with significant consequences for match rates on the platform. Table 3 demonstrates a strong negative correlation between the number of bids a job posting receives and the probability of the employer hiring a worker. We regress in this table the cross-sectional probability that a project gets awarded to a worker on the number of competitive bids a project receives, controlling for a job posting’s budget category. We include a job category fixed effect to control for unobserved heterogeneity across different skill categories (PHP, Python or Data entry). The regression specification is described in Equation 18. Figure 11 provides a binscatter visualization of this regression framework. Appendix Figure B.2 further splits this binscatter by job category and by budget group, showing that the probability that a job gets awarded is lower for skill-intensive jobs (Python and PHP) than for low-skill jobs (Data entry).²⁰ Projects posted in lower budget groups (below USD100) are also more likely to find a match, but this probability decreases as well with the number of competitive bids received for the project, and in the same magnitude.

$$\text{Awarded}_j = \beta_0 + \beta_1 \times \text{BidsNumber}_j + \sum_k \beta_k \times \text{Budget}_{jk} + \gamma_{\text{JobCat}[j]} + \varepsilon_j \quad (18)$$

where awarded_j is a dummy variable equal to 1 if project j is awarded to a worker, and 0 otherwise, BidsNumber_j is the number of bids project j receives, Budget_{jk} represents a categorical variable for different budget groups for project j , with k indexing the groups (2 budget groups, in USD: “below USD 100” and “above USD 100”). $\gamma_{\text{JobCat}[j]}$ is a fixed-effect for the job category under which project j is posted (either Python, PHP and Data entry) and ε_j is the error term for project j , assumed to be normally distributed.

²⁰PHP jobs see a relatively lower decrease in the probability of being awarded as a function of the number of competitive bids.

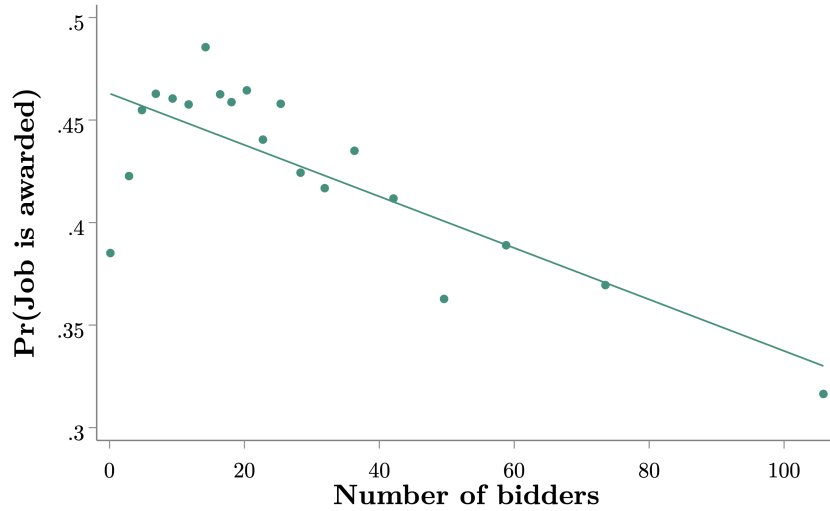


Figure 11: The probability that employers pick a worker decreases with competition

Notes: This binscatter figures reports the coefficient from equation (18). It displays a strong negative correlation between the number of competitive bids submitted to a posted job and the probability that the prospective employer picks a winner.

Table 3: The probability that employers pick a worker decreases with competition

	Dependent Variable: Pr(project is awarded)		
	(1)	(2)	(3)
Number of Bids Per Job Posting	-0.00114*** [0.00011]	-0.00126*** [0.00012]	-0.00088*** [0.00012]
Budget > 100			-0.06885*** [0.00706]
Job Category FE		✓	✓
Constant	0.4596*** [0.0046]	0.463*** [0.0046]	0.4987*** [0.00592]
F-Stat	100.3	118.7	105.9
Dep. Var.: Mean	0.4269	0.4269	0.4269
Dep. Var.: Std. Dev.	0.4946	0.4946	0.4946
N Obs.	24,681	24,681	24,681

Notes: * ($p < 0.1$), ** ($p < 0.05$), *** ($p < 0.01$). Standard errors are clustered at the FE level. This table reports the coefficient from equation (18). We find a strong negative correlation between the number of competitive bids submitted to a posted job and the probability that the prospective employer picks a winner. The average probability that an employer effectively outsources the posted job to a worker strongly decreases with the number of bids received.

4.5 Relationship between Reputation and Efforts

As workers build their reputation on the platform (i.e. as they get more employer reviews post-completion of job tasks), we observe a decrease in the effort they put into their job applications. Table 4 shows a strong negative correlation between the number of reviews a worker has received on the platform and the time they spend on their applications (i.e. their “effort”). We regress in this table a worker’s effort, as measured by the timing of their bid application (time between their first click on a job posting and submission of their job proposal) on worker’s reputation, controlling for a job posting’s category (Python, PHP or Data Entry jobs) and budget category. All regressions include a worker fixed effect, as well as workers’ country fixed effect to control for country-specific unobserved heterogeneity.

The regression specification is described in Equation (19). Figure 12 provides a bin-scatter visualization of this regression framework.

$$\begin{aligned} \text{Effort}_{ij} = & \beta_0 + \beta_1 \times \text{Reviews}_i + \sum_k \beta_k \times \text{JobCategory}_{jk} \\ & + \sum_m \beta_m \times \text{BudgetGroup}_{jm} + \gamma_i + \gamma_{\text{country}[i]} + \varepsilon_{ij} \end{aligned} \quad (19)$$

where Effort_{ij} is our worker-job specific effort measure, which computes the time it takes for worker i to customize their bid application for job j , Reviews_i is the number of workers freelancer i has received from previous employers (reputation), JobCategory_{jk} represents the dummy variables for different job categories for project j , with k indexing the categories, BudgetGroup_{jm} represents the dummy variables for different budget groups for project j , with m indexing the groups. γ_i and $\gamma_{\text{country}[i]}$ represent the fixed effects for freelancers i and their countries, respectively. ε_{ij} is the error term for each observation of freelancer i on project j , assumed to be normally distributed.

This finding suggests that as a worker’s observable characteristics (reputation) become more predictive of their match quality, they can afford to spend less effort on providing detailed match-specific information (a “signal” of job-specific fit). Employers, in turn, rely more heavily on these observable characteristics when evaluating workers with established reputations.

In Figure 13, we split the binscatter Figure 12 into budget groups. Higher-paying jobs necessitate higher signaling efforts from Freelancers, even as their reputation improves. Appendix Figure B.5 further splits the binscatter Figure 12 into job categories

and aggregated budget groups.

Table 4: Workers exercise less effort (write and submit their bids faster) as their reputation improves

	Dependent Variable: Bid Timing (Effort)		
	(1)	(2)	(3)
Worker's Number of Reviews	-.00421*** [.0001]	-.00421*** [.0001]	-.00407*** [.0001]
Job Category			
Data entry (baseline)		-	-
PHP		.00284 [.01036]	-.02109** [.01047]
Python		.0552*** [.01366]	.02728* [.01405]
Budget Group (USD)			
10-30 (baseline)			-
30-250			.1309*** [.00561]
250-750			.2199*** [.00633]
750+			.3622*** [.0096]
Constant	1.774*** [.01482]	1.769*** [.01748]	1.604*** [.01823]
Worker's FE	✓	✓	✓
Worker's Country FE	✓	✓	✓
F-Stat	1,725	583.5	564
Dep. Var.: Mean	1.162	1.162	1.143
Dep. Var.: Std. Dev.	1.623	1.623	1.597
N Obs.	5.7e+05	5.7e+05	5.1e+05

*Notes:** ($p < 0.1$), ** ($p < 0.05$), *** ($p < 0.01$). Standard errors are clustered at the FE level. This figure is a binscatter visualisation of Equation (19). It displays the cross-sectional relationship between bid timing (job application “effort”) and the a freelancer’s reputation as measured by their number of reviews from past employers (see regression table 4). The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. The number of worker’s reviews captures the working history of a freelancer (more reviews means a more established history of work on the platform). A more established work history places workers at the forefront of bid rankings when they apply for jobs. This plot shows that workers bid faster as their reputation improves, and thus exerts less efforts to signal their job-specific fit.



Figure 12: **Workers exercise less effort as their reputation improves**

Notes: This figure is a binscatter visualisation of Equation (19). It displays the cross-sectional relationship between bid timing (job application “effort”) and the a freelancer’s reputation as measured by their number of reviews from past employers (see regression table 4). The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. The number of worker’s reviews captures the working history of a freelancer (more reviews means a more established history of work on the platform). This plot shows that workers bid faster as their reputation improves, and thus exerts less efforts to signal their job-specific fit.

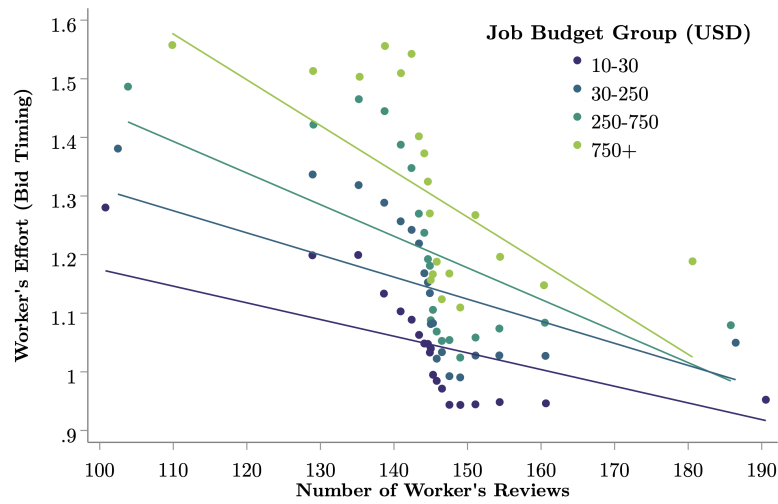


Figure 13: **Workers exercise less effort as their reputation improves**

Notes: This figure displays the cross-sectional relationship between bid timing (job application “effort”) and the a freelancer’s reputation as measured by their number of reviews from past employers (see regression table 4). We split binscatter 12 into budget groups. Higher-paying jobs necessitate higher signaling efforts from Freelancers, even as their reputation improves. The bidding time is measured as the time-lapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. The number of worker’s reviews captures the working history of a freelancer.

4.6 Does Signal Quality Predict Employers' Choices?

We eventually want to test whether employers respond to our measure of signal quality (“content fit”). To perform this test, we set up a multinomial logit to predict employer choices based on our LLM measure of signal quality (described in Section 4.1), bids and workers characteristics. This regression framework is a reduced-form version of the employer’s discrete choice logit model laid down in Section 3.

Equation 20 describes employer j ’s utility from awarding their job task to worker i :

$$U_{ij} = \beta_1 \text{Signal Quality}_{ij} + \beta_2 \text{Bid}_{ij} + \beta_3 \text{Effort}_{ij} + \beta_4 \text{Worker Characteristics}_i + v_{ij} \quad (20)$$

where $\text{Signal Quality}_{ij}$ is worker-project match specific signal quality extracted from the LLM (post logit transformation), Effort_{ij} is a measure of application-specific effort (bid timing, described in Section 4.1), Bid_{ij} is the bid amount submitted by worker i for the job j , $\text{Worker Characteristics}_i$ is worker i ’s observable characteristics on the platform (country group and reputation).²¹ v_{ij} is an extreme-value type 1 error term. The employer’s outside option is to not pick any worker from the platform.

We run these regressions for two job categories for which we have carried LLM evaluations of job application contents: PHP and Python. We only keep one budget group for now (USD 30-250), which is the modal budget for posted projects. For this budget category, PHP has 59,966 bids and 2,358 projects, 43.98% of which have winners. Python has 10,628 bids and 938 projects, 35.91% of which have winners.

Table 5 shows that match-specific signal quality matters for employer choices. Our measure of content fit is strongly positively correlated with the probability that a worker is selected for a project, both for PHP and Python jobs. Employers “care” about 1 standard deviation of content fit 17% as much as they care for a 1 standard deviation of bid amount (\$66) for PHP (Columns 1 and 2). Employers “care” about 1 standard deviation of content fit 59% as much as they care about a 1 standard deviation of bid amount (\$70.4) for Python projects (Columns 3 and 4). Higher bid amounts are negatively correlated with a probability of winning an auction (significant at the 1% level across all specifications). Moreover, employers respond to effort (second row of the table in Columns 2 and 4). This further motivates the modeling assumption that effort is observed and relevant

²¹The variable **Reputation** captures a worker’s reputation, as measured by their star rating and number of reviews. We create 5 groups: “rookies” (if the freelancer is new on the platform and do not have an established reputation yet), “bad” (if a worker’s star rating is below 4 on a 0-5 scale), “good (below 50 reviews)” (if a worker is above 4 stars but has less than 50 employer reviews), “good (50-200 reviews)” (if a worker is above 4 stars and has between 50 and 200 employer reviews) and “good (above 200 reviews)” (if a worker is above 4 stars and has more than 200 employer reviews). Very few people in our sample have less than a 4-star rating (1.3% for PHP and 1.7% for Python).

to the employers' choice and matching outcomes.

Table 5: **Logit Regressions - Does Signal Quality Predict Employers' Choices?**

	Dependent Variable: Pr(win)			
	PHP		Python	
	(1)	(2)	(3)	(4)
Bid Amount	-0.01326*** [0.00061]	-0.0133*** [0.00061]	-0.01218*** [0.00101]	-0.01214*** [0.00101]
Bid Time (Effort)		0.05081** [0.02357]		0.07829*** [0.02679]
Content Fit (Signal Quality)	0.06685*** [0.0165]	0.05872*** [0.01687]	0.1973*** [0.02643]	0.1742*** [0.0274]
Country Category				
East Asia	-2.798*** [0.2332]	-2.873*** [0.2359]	-2.445*** [0.2865]	-2.634*** [0.2948]
South Asia	-3.213*** [0.2122]	-3.311*** [0.2175]	-2.446*** [0.2419]	-2.665*** [0.2548]
English Speaking	-2.645*** [0.2563]	-2.742*** [0.2605]	-2.07*** [0.297]	-2.294*** [0.3084]
Europe	-2.281*** [0.2271]	-2.377*** [0.2316]	-1.904*** [0.2596]	-2.138*** [0.2734]
Other	-2.834*** [0.2315]	-2.94*** [0.2369]	-2.275*** [0.2578]	-2.5*** [0.2706]
Reputation				
bad (below 4 stars)	0.195 [0.5402]	0.2329 [0.5405]	-0.583 [1.027]	-0.5815 [1.029]
good (below 50 reviews)	1.067*** [0.2069]	1.107*** [0.2078]	0.9821*** [0.216]	1.079*** [0.2193]
good (50-200 reviews)	1.718*** [0.2062]	1.766*** [0.2076]	1.185*** [0.2284]	1.319*** [0.2342]
good (above 200 reviews)	2.258*** [0.2092]	2.306*** [0.2105]	1.61*** [0.2519]	1.744*** [0.2573]
Number of Bids	59,966	59,966	10,628	10,628
Number of Projects	10,628	10,628	938	938
Dep. Var.: Mean	0.03783	0.03783	0.08466	0.08466

Notes: * (p<0.1), ** (p<0.05), *** (p<0.01). **Country Category** is a categorical variable capturing a worker's geographic region (East Asia, South Asia, Europe, "English Speaking" and Other). "English Speaking" countries are the US, Canada, UK, Australia, New Zealand. The variable **Reputation** captures a worker's reputation, as measured by their star rating and number of reviews. We create 5 groups: "rookies" (if the freelancer is new on the platform and do not have an established reputation yet), "bad" (if a worker's star rating is below 4 on a 0-5 scale), "good (below 50 reviews)" (if a worker is above 4 stars but has less than 50 employer reviews), "good (50-200 reviews)" (if a worker is above 4 stars and has between 50 and 200 employer reviews) and "good (above 200 reviews)" (if a worker is above 4 stars and has more than 200 employer reviews).

Note that these results are *descriptive*, in the sense that employers do *not* have true preferences over signals. They have preferences over what they are learning from the signals – that is match quality. To recover employers’ preferences, we thus need to model how employers form beliefs over match quality – which Section 3 does.

4.7 Summary of Reduced-Form Evidence

Our empirical evidence highlights a tension between market thickness and information provision in online labor markets. As markets become thicker and competition intensifies, workers are discouraged from investing substantial effort in crafting high-quality, personalized applications. This results in a high volume of generic, low-effort bids, making it difficult for employers to identify the most suitable candidates. Our descriptive evidence suggests this contributes to low match rates in this digital marketplace. The reduced information provision by workers can be seen as a negative externality of increased competition in thick markets, threatening the efficiency gains that market thickness is expected to deliver. These findings underscore the importance of information transmission in the matching process and the need for mechanisms that encourage more informative communication between workers and employers.

5 Conclusion

This paper explores market thickness, competition, and information provision in online labor markets. Our work highlights a fundamental tension in global digital marketplaces: while increased market thickness can potentially lead to more high-value matches, it also intensifies competition among workers, disincentivizing them from providing costly match-specific information to employers. This tension, coupled with the presence of attention-constrained employers, can result in an equilibrium characterized by hasty and generic job applications and low job-filling rates.

To formally investigate this phenomenon, we develop a model of equilibrium information provision in which attention-constrained employers evaluate global workers based on their observable characteristics, price bids, and costly signals of match quality. Workers provide match-specific costly information to their potential employers. Our model captures the key features of online labor markets, such as the one-shot nature of job postings, the importance of information provision in screening, and the role of recommender systems in shaping employers’ consideration sets. By embedding a signaling model into a scoring auction framework, we are able to analyze the impact of congestion

on workers’ incentives to provide informative signals and the resulting market outcomes.

Our empirical analysis using data from `Freelancer.com` provides suggestive evidence for the mechanisms highlighted in our model, including market congestion and its effects on workers’ effort and application quality. We find that, on average, employers receive a large number of applications per job posting, indicating a high level of competition among workers. However, despite this high volume of applications, we observe that a significant proportion of these applications exhibit low effort and low levels of customization, as measured by the time spent on the application, the length of the proposal, and the text-analysis of the job posting and job applications by a LLM. Furthermore, we document that workers tend to submit applications quickly after job postings, suggesting that they may be dedicating little time to reading the job requirements before applying. These findings, along with the low job-filling rates and bid-acceptance rates observed in our sample, support the mechanisms highlighted in our model. Intense competition and the presence of congestion appear to discourage worker from investing substantial effort in crafting high-quality, personalized applications, as they internalize their low chances of being considered. Consequently, employers often face a large pool of undifferentiated and uninformative applications, making it challenging to identify the most suitable candidates for their jobs. Low matching rates suggests that workers may not be providing sufficient information to convince employers to hire them.

These findings have important implications for the design and regulation of online labor markets. Platform designers could consider implementing mechanisms that encourage workers to provide more informative and targeted applications, even in the presence of congestion. This could involve refining recommender systems to prioritize applications with higher-quality signals or introducing reputation systems that reward workers for providing valuable information to employers. Additionally, employers could be incentivized to provide more detailed job descriptions and requirements, which could allow workers to better customize their applications.

Our work also contributes to the broader literature on market design, highlighting the importance of considering information provision and congestion in the context of two-sided markets. The insights from our model and empirical analysis can be applied to other settings where agents compete for matches and face the trade-off between providing costly information and the likelihood of being considered, such as college admissions, online dating, and housing markets.

Furthermore, our paper demonstrates the potential of large language models (LLMs) in enabling the empirical estimation of communication games. By leveraging LLMs to quantify the information content of job applications, we open up new avenues for re-

search at the intersection of information economics, market design, and natural language processing. This approach can be extended to other domains where the analysis of unstructured text data is crucial for understanding strategic interactions and market outcomes.

Looking ahead, our model provides a framework for investigating potential counterfactuals that could help alleviate the negative externalities arising from market thickness and competition. One avenue for future research is to simulate an equilibrium in which workers behave as if they faced no competition. This counterfactual would allow us to assess the extent to which the negative externality of market thickness can be mitigated without limiting employers' choice sets. Another direction is to simulate equilibria with varying levels of market thickness, enabling us to identify the threshold at which the negative externality begins to outweigh the benefits of increased market size. Finally, simulating equilibria with different signaling technologies, such as the introduction of LLMs that may erode signal informativeness, can provide insights into the role of technology in shaping market outcomes and the potential consequences of relying on automated systems for screening and matching.

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Appendices

A Distinctions with Spence (1973) signaling model

There are three key distinctions between our signaling model and the canonical [Spence \(1973\)](#) signaling model, the analysis of which is typically split between studying pooling equilibria, i.e. agents of different types make the same observable choice, and separating equilibria, i.e. agents of different types make different observable choices.

The first key distinction is that agents are not self-sorting into payoffs (wages in the canonical Spence model) depending on their observable actions (educational choice in the canonical Spence model). Instead, under our setting, agents are competing for a discrete choice on behalf of the employer. Put differently, the employer is not setting a wage that is conditional on observable actions, but instead they are choosing the bid that maximizes their expected utility, i.e. the scoring rule, and bidders set their own wages as part of their bid.

The second key distinction is that, due to the continuous noise e_i being added to each bid, there will be no point masses in the equilibrium distribution of writing quality regardless of whether bidders are pooling on a choice of effort. Thus, we will not have “pooling” in the canonical Spence sense in this model. Moreover, workers’ ability is not their private information and workers do not know their types. They choose an effort knowing it will produce some type of signal on quality, but do not know their project-specific match quality, which is their type under our framework.

The third key distinction is the presence of another observable dimension of actions, i.e. b_i . Employers do not just observe the costly signal, and they know there is another dimension of unobservables, i.e. v_i , that explains the bid transfer and quality signal pairs they observe.

In sum, this model does not fall neatly into the canonical discussions of pooling and separating equilibria.

B Additional Reduced-form Results

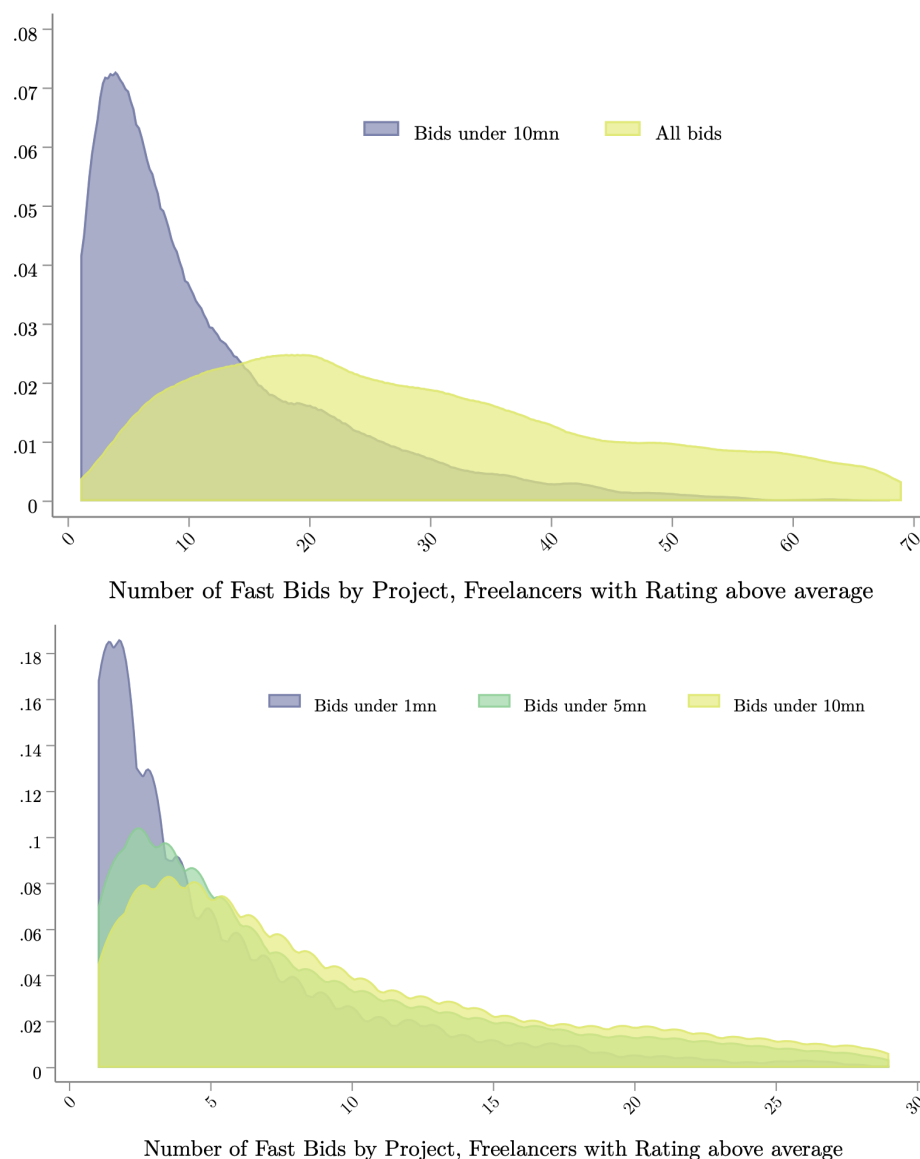


Figure B.1: Employers have many fast bids to sort through

Notes: These figures plot the distribution of arrival times for competitive bids send to posted job offers. We focus on the sample of workers (“freelancers”) that display a star rating higher than average (“serious” workers). For each project posted on the platform between January 2019 and March 2020 in our sample, we look at the number of bids received from established freelancers, and the timing of those bids. The top figure plots the distribution of bids received under 10 minutes and compares it to all received bids. The bottom figure splits fast bids from established freelancers into 3 categories: bids submitted under 1 minute, under 5 minutes and under 10 minutes. Bids from workers with a good rating history tend to be written quickly.

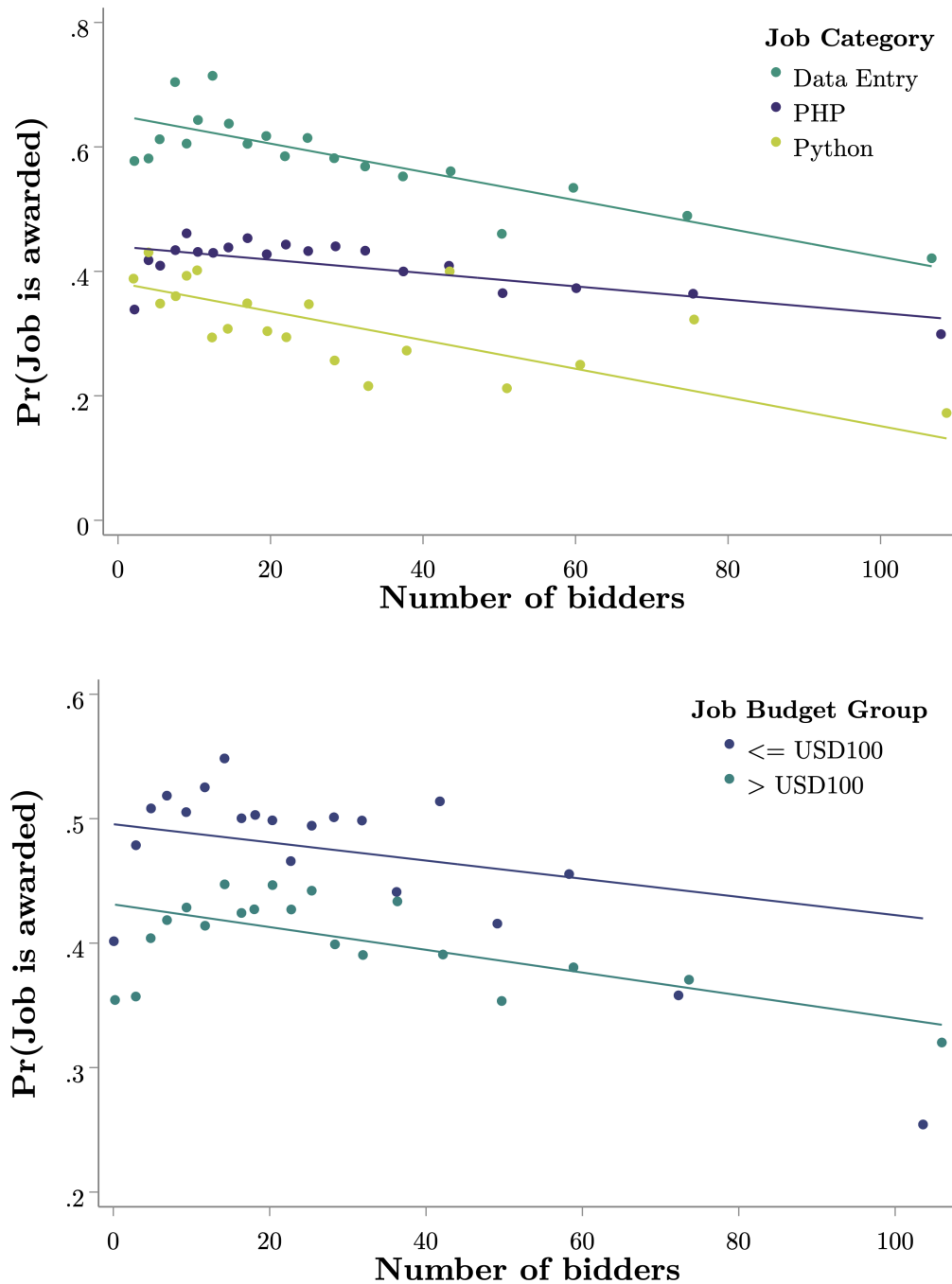


Figure B.2: The probability that employers pick a worker decreases with the number of competitive bids they receive

Notes: This binscatter figures report a strong negative correlation between the number of competitive bids submitted to a posted job and the probability that the prospective employer picks a winner. The average probability that an employer effectively outsources the posted job to a worker strongly decreases with the number of bids received. 60% of projects posted in 2019 did not get awarded to a worker.

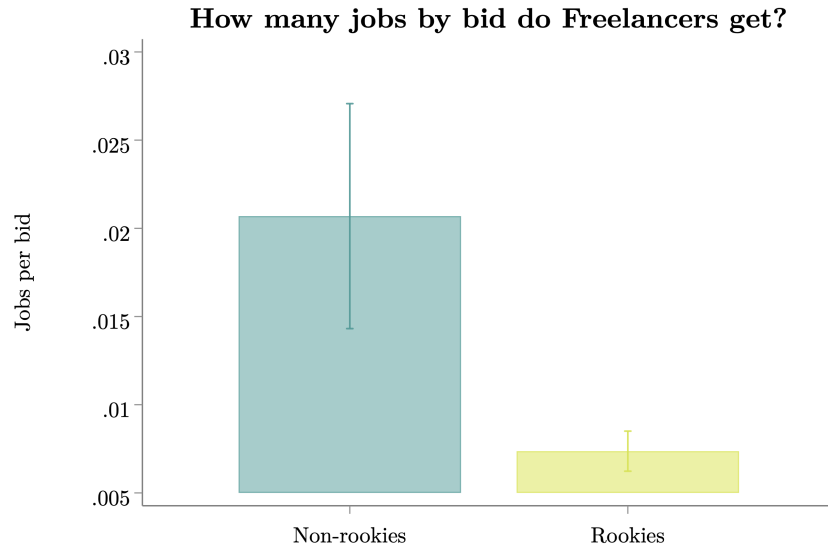


Figure B.3: Probability of Obtaining a Job

Notes: This figure displays the ratio of jobs obtained per bid submitted across a 5% random sample of workers (“freelancers”) working on the platform between 2016 and 2019. The conversion rate from bid to job is extremely low, and even “non-rookies” get on average only 0.02 jobs per bid. Rookies are defined as all workers who are active on the platform but have not yet received a first milestone payment at the time of bidding. Non-rookies are workers that have received at least one payment at the time of bidding.

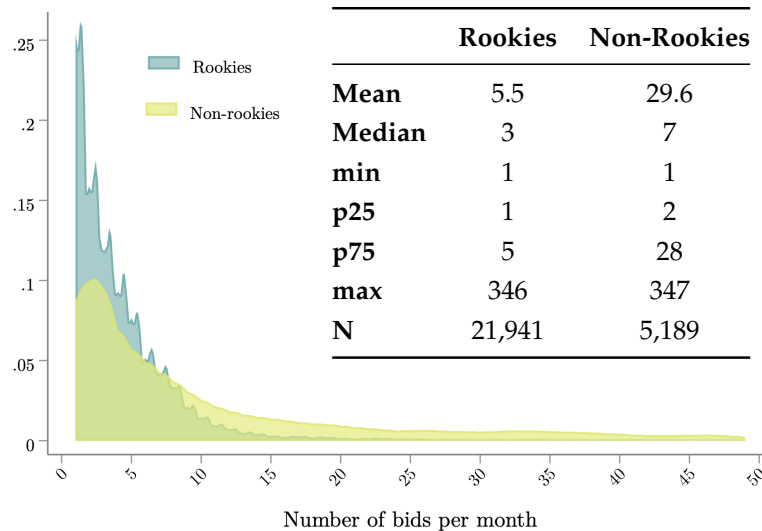


Figure B.4: Established Workers Bid More

Notes: “Non-rookie” workers submit many more bids per months than “rookies”, suggesting that the optimal strategy to obtain jobs in this congested marketplace is to bid a lot. Rookies are defined as all workers who are active on the platform but have not yet received a first milestone payment at the time of bidding. Non-rookies are workers that have received at least one payment at the time of bidding.

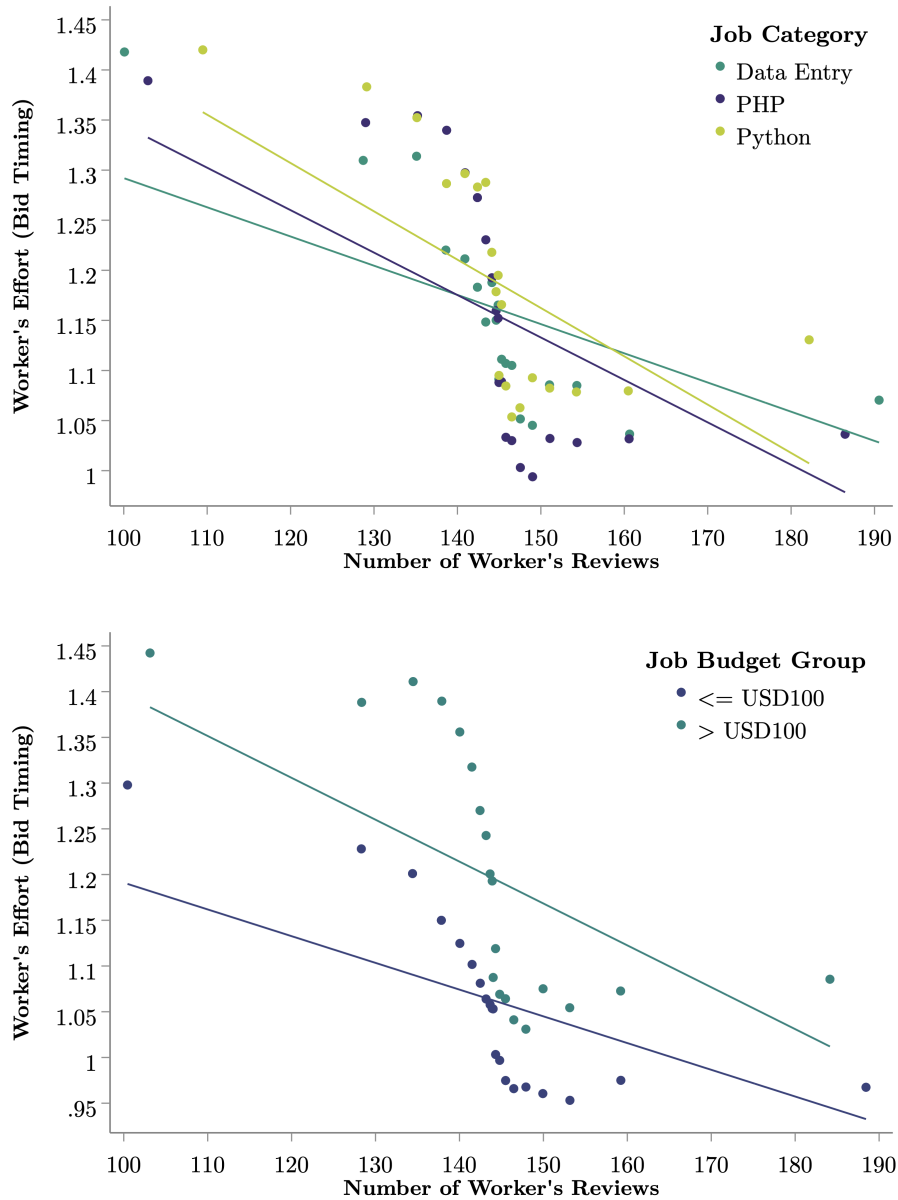


Figure B.5: Workers exercise less effort (write and submit their bids faster) as their reputation improves

Notes: This figure displays the cross-sectional relationship between bid timing (job application “effort”) and the a freelancer’s reputation as measured by their number of reviews from past employers (see regression table 4). The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. The number of worker’s reviews captures the working history of a freelancer (more reviews means a more established history of work on the platform). A more established work history places workers at the forefront of bid rankings when they apply for jobs. This plot shows that workers bid faster as their reputation improves, and thus exerts less efforts to signal their job-specific fit. The top graph splits the original binscatter 12 into job-specific categories, showing that this relationship holds independently of the skill-intensivity of the traded task (Pyhon jobs are the most skill intensive, followed by PHP, and in third Data entry). The bottom graph splits binscatter 12 into budget groups. Higher-paying jobs (above USD 100) necessitate higher signaling efforts from Freelancers, even as their reputation improves.

C Measuring Writing Effort and Writing Quality

C.1 LLM-Based Measuring Methods: An Overview

The limitation of traditional NLP approaches is that they are not well-suited for measuring “text fit.” To address this, we propose a new measure of writing quality that relies on a Large Language Model (LLM) to evaluate both the employer’s job descriptions and the workers’ submitted job application materials. Our measures aim to penalize job applications that, despite being well-written, are generic or recycled from other applications, and conversely, to reward letters that, although potentially less polished, directly address the specifics of the job description. This methodology is designed to assess the degree to which each proposal is tailored to and appropriate for the respective job posting.

This approach illustrates the innovative application of LLMs in quantifying aspects of text data that were previously beyond the reach of conventional analytical methods.

To measure the signaling fit of a worker’s job application, we rely on a “Yes/No” evaluation of a job application’s fit to a specific job posting by querying the ChatGPT API. We then use a posterior probability extraction method to recover the probability that a “yes” was issued by the LLM.

This approach involves asking the LLM to determine whether a bid proposal appears well-suited and specifically tailored for the job posting through a simple yes/no question. This binary assessment focuses on evaluating the fit and customization of the proposal relative to the job requirements.

Upon receiving a yes/no answer from the LLM, we extract the model’s posterior probability of a “yes” response, which serves as the score for each proposal. This score reflects the LLM’s confidence in the proposal’s suitability and customization for the job. To measure the posterior probability of a “yes” answer, we leverage the LLM’s ability to assess the likelihood of certain outcomes based on the input data. This process involves querying the LLM with a prompt that includes the bid proposal and the job description, asking it to evaluate the suitability and customization of the proposal in a yes/no format.

The LLM, based on its internal neural net models and the training it has received, can then evaluate the input data and provide a “yes/no” answer. The ChatGPT API endpoint supports returning probabilities for its responses, which can be directly requested along with the “yes/no” answer.

The extracted posterior probability for a “yes” answer then serves as a quantifiable measure of the proposal’s alignment with the job description. This numerical score can be used to rank proposals, analyze the effectiveness of the bidding process, and

potentially guide improvements in how freelancers craft their proposals. The advantage of this method lies in its simplicity and the direct measure of fit and qualification it provides.

C.2 Prompts Submitted to the LLM for Bids Evaluation

We provide here an overview of the prompt used via the Chat GPT's API to measure content fit.

We feed the API with an employer's job posting and prompt it to role-play as the firm posting the task. We then feed the LLM with the associated workers' bid applications (without any other information about workers) and prompt it to score the writing based on a judgment of how well the freelancer writing the cover letter will do on the task described.

Our prompt to the LLM's API follow the following structure:

- You are the employer who posted the job
- You can only answer "Yes" (hire) or "No" (don't hire)
- It is never optimal to answer yes or no all of the time, and instead you want to randomize
- Randomize by choosing a weighted coin, but only report the result of the coin flip
- Criteria for the choice
 - Application matches the details of the job post
 - Applicant seems qualified and to understand the details of the job
 - Look out for cheap talk like "I have read your job post carefully"
 - Look out for attention checks
 - Value applicants that propose details of how they would do the job
- Few shot optimization is key

C.3 Example of Job Posting and Job Applications

We provide here an example of the LLMs rating measure, based on posterior probability extraction on a yes/no question about job application fit to job postings.

The **employer's job posting** is as follows:

I would need a python script to do pre-processing the data from a MySQL table. The table contains text messages from the users. (sample attached: messages.sql) The script need to take the text messages and filter it based on the the dictionary table (dict.sql) If you are using tokenizer to find word, please note that in this dataset: 1) punctuation itself is also considered a word 2) it is not case sensitive, e.g. china = China Therefore for a text "how I loathe thy tasty good" if only the word thy is NOT in the dictionary, the filtered result should be "how i loathe tasty good." The filtered result should be stored back into MySQL table.

Here are two different job applications for this posting:

Application 1

Hi, Dear Employer! I am really interested in your project. I have enough experience in Python, C/C++, java programming. Especially I am an expert in NLP and Machine Learning. I am 100% sure I can satisfy your requirements perfectly. Thank you and best regards!

Application 2

Hi, I'm expert in python and NLP, also I've done similar project for what you need (pre-processing emails, creating word dictionary then creating word vectors for each email) for spam classifier, I've also good experience with sql databases, feel free to check sample of my work <https://github.com/XX/Email-spam-classifier-python>, and to discuss any further details.

The first application was written and submitted in 30 seconds (bidding time measure, or "effort"), while the secon one was written in 5 minutes. The "content fit" (signal quality) measure from the LLM is 0.26 on the first application (on a 0-1 scale), which becomes 4.07 after logit transformation, while the second application got a score of 0.98 (-1.0 after logit transformation).

D Market Description - Additional Figures

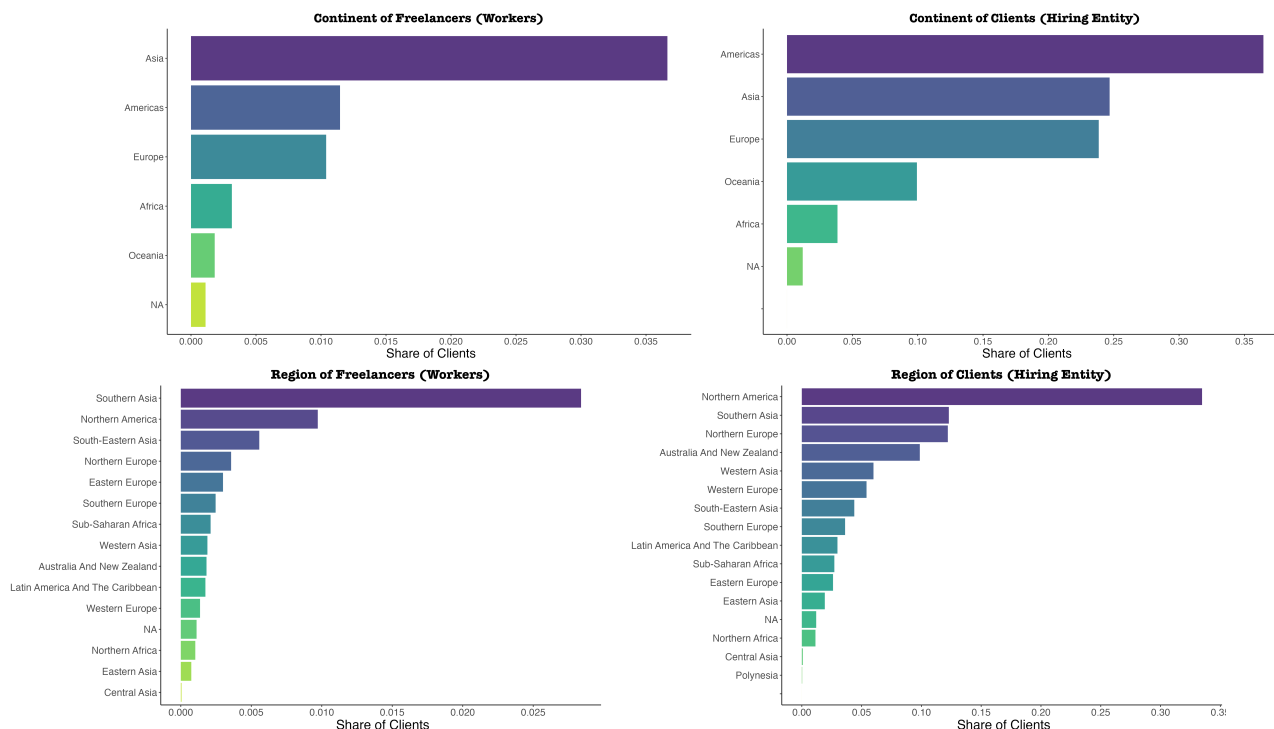


Figure D.1: Continents and Regions of Clients and Freelancers

Notes: These figures plot the share of workers (“freelancers”) and employers (“clients”) by continent and world regions. The two top figures show the geographical distribution of workers (left) and employers (right) by continent, while the two bottom figures show their spatial location across regions. Firms on the platform are largely from the Global North, while freelancers on the platform are mostly from the Global South. Most freelancers come from Asia, and more particularly Southern Asia, while most clients looking to outsource service tasks are from Northern America and Europe.

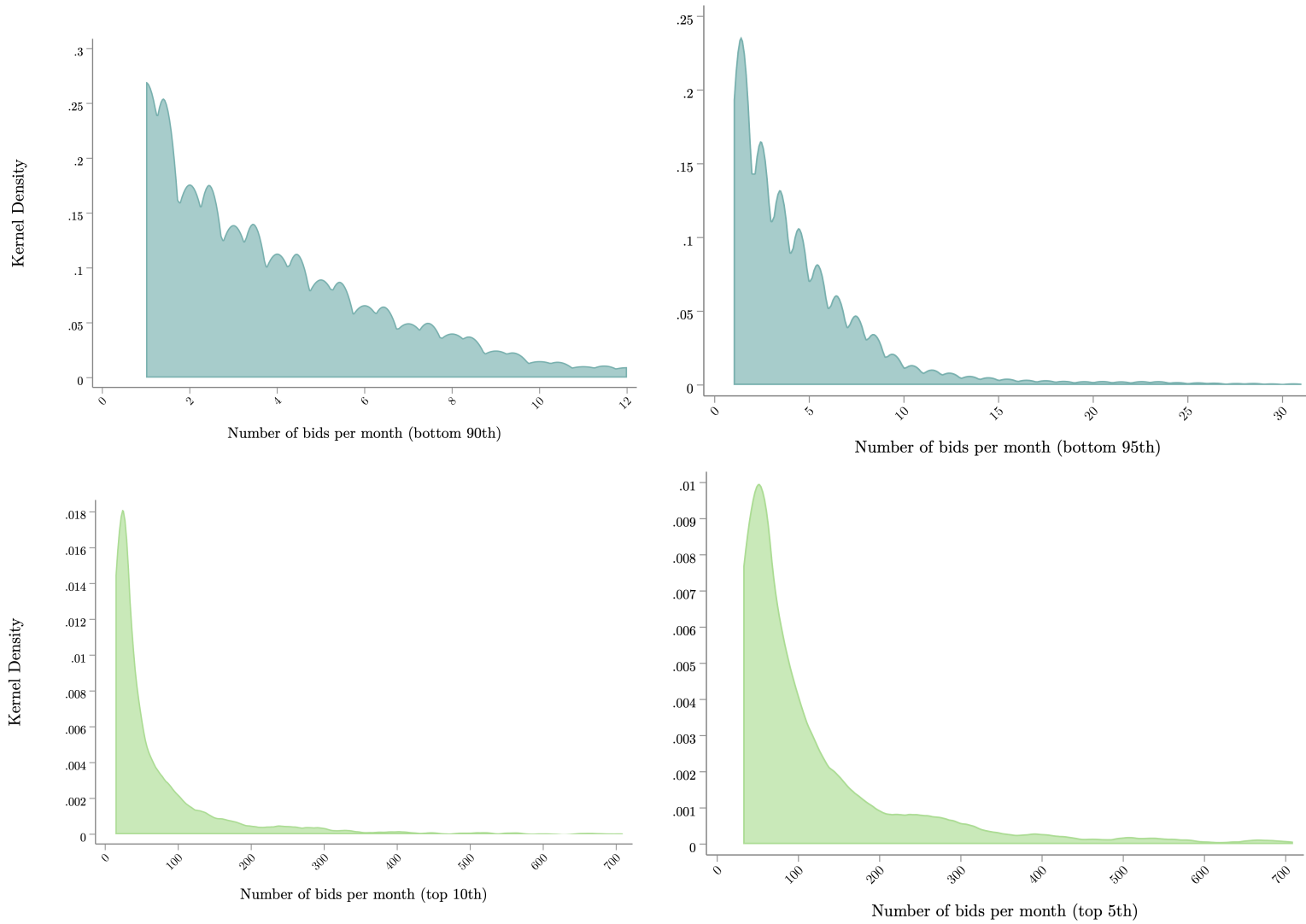


Figure D.2: Freelancers Are Paying for Memberships with High Bid Budgets

Notes: These figures plot the distribution of bids submitted by workers on the platform. The top two figures display the bottom 90th and 95th percentiles of workers, while the two bottom figures focus on the top 10th and 5th percentiles. Distributions are over workers. Workers at the top bid a lot. This provides evidence that workers pay for membership plans that override the free-plan's bid threshold and allow them to bid a lot (see Appendix D.3 for a detail of the plans).

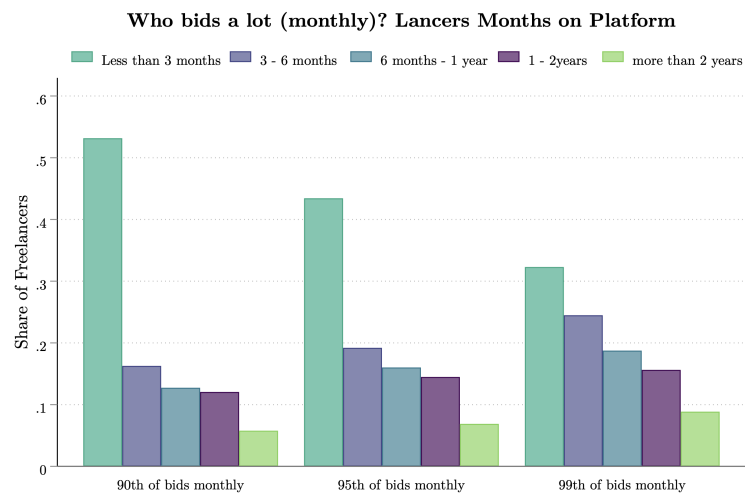
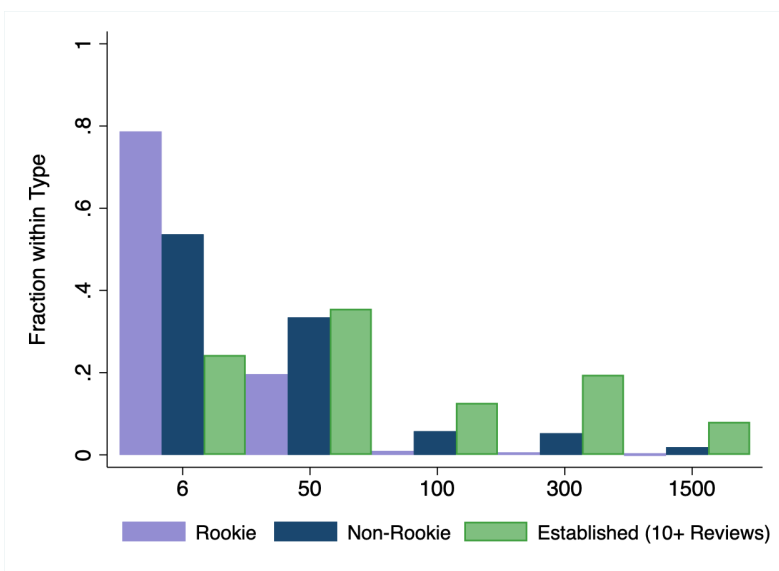
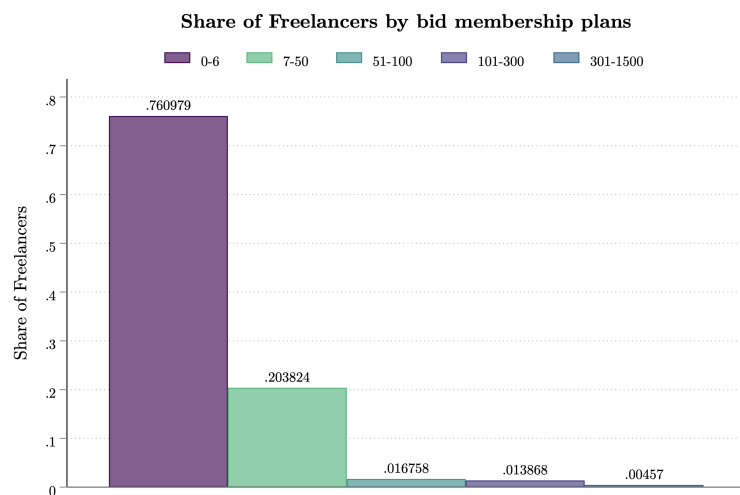


Figure D.3: Workers are Paying for Memberships that Permit to Submit more Bids

Notes: These figures display the distribution of workers (“freelancers”) across bid membership plans (see Appendix B5 for a detail of the plans). Memberships allow workers to bid more than the free membership plan. There are currently 6 membership plans. The free membership plans allows to submit 6 bids per month, the second tier up to 50, and the three following plans up to 100, 300 and 1500. The top left figure shows that 20% of freelancers select to pay and submit up to 50 bids a month. Higher limit plans display smaller share of workers. The top right figure further decomposes membership payments by worker seniority type. “Rookies”, i.e. freelancers that never received a payment for a job, predominantly select into the first two tiers of memberships. More experienced freelancers tend to pay for higher limit plans. The more established the freelancers, the more likely they are to pay for plans that provide very high bid thresholds (including the 300 monthly bid and 1500 bid plans). The bottom figure further splits freelancer by the length of time they spent on the platform, and look at the number of bids they submit every month. The longer freelancers have been working on the platform, the more likely they are to bid a lot.

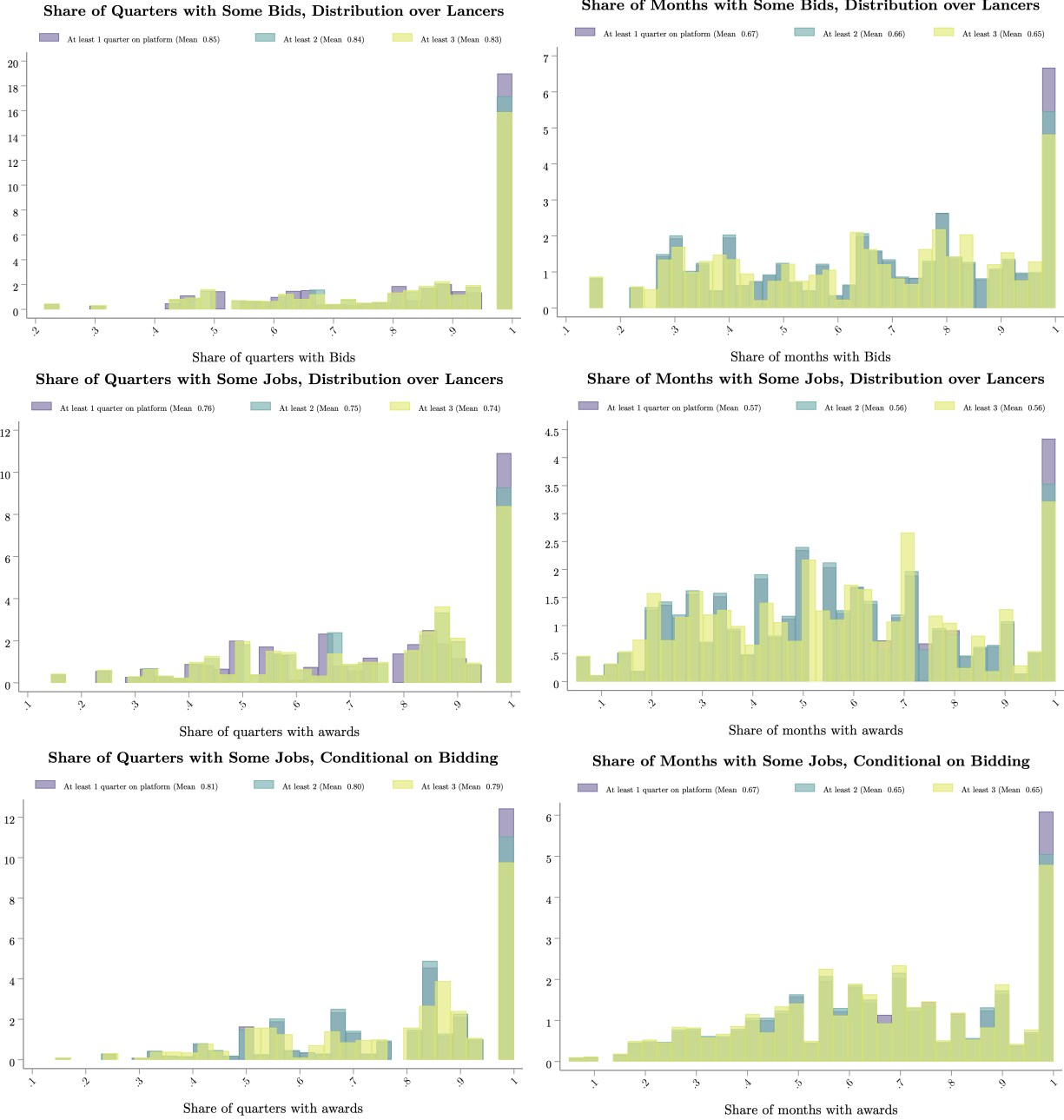


Figure D.4: Density Plots - Time Distribution of Bids and Jobs

Notes: These barplots display the time distribution of bids and jobs for workers (“lancers”) on the platform over the period january 2016 to december 2019. All figures on the left are at the quarter level and those on the right are at the monthly level. We split the bins for three groups of freelancers: i) those who have been on the platform for at least 1 quarter (in purple), ii) at least 2 quarters (green) and iii) at least 3 quarters. The first two top graphs plots the share of time during which freelancers submit some bids. The middle two graphs show the share of time during wich freelancers have some jobs on the platform. The last two graphs show the share of time during which freelancers have some jobs on the platform, but conditional on bidding.

E Data Appendix

The following list details the main variables we use from Freelancer . com for this project.

Identifiers include:

- Project ID
- Bid ID
- Client ID
- Freelancer ID

Client/Freelancer Information:

- Did the freelancer pay for upgrade
- Is the client/freelancer a new or returning user (Yes/No)
- Client's number of past projects
- Freelancer's number of past bids
- Number of past reviews and corresponding scores (for client/freelancer)
- Country and state (for clients/freelancers)
- Client's past total spend

Bidding Information:

- Bid timestamps
- Bid price and job application text
- Actions related to a bid: whether it was selected, final payment, reviews

Project information:

- Project upgrade fees charged
- Project milestones
- Project skills
- Milestone actions (release, reject, update, cancel) and timestamps
- Project budget with respective currency noted (or all in US dollars if possible)
- Revenue earned per project
- Indicators for which bidders clients chat with, including timestamps and character counts.
- Project success metrics and feedback provided.

F Platform Details Appendix

Table F.1: Job Categories and Associated Skills

Job Category	Skill Examples	Number of Skills
Websites, IT & Software	PHP, HTML, JavaScript, Software Architecture	15283
Design, Media & Architecture	Graphic Design, Website Design, Photoshop	15244
Mobile Phones & Computing	Mobile App Development, Android, iPhone	1752
Data Entry & Admin	Data Entry, Excel, Data Processing	1698
Engineering & Science	AutoCAD, CAD/CAM, Engineering	2394
Freight Shipping & Transportation	Dropshipping, Containerization, Import/Export	38
Business, Accounting, Human Resources & Legal	Business Analysis, Accounting, Finance	1317
Product Sourcing & Manufacturing	Product Design, Product Sourcing, Supplier Sourcing	161
Sales & Marketing	Internet Marketing, Social Media Marketing, Sales	3570
Translation & Languages	English (US) Translator, Spanish Translator, French Translator	574
Writing & Content	Content Writing, Article Writing, Copywriting	3858
Artificial Intelligence	Large Language Models (LLMs), Bot Development	136
Education & Tutoring	English Tutoring, Language Tutoring, Chemistry Tutoring	28
Health & Medicine	Food Safety, Counseling Psychology, Public Health	8
Telecommunications	Physical Site Survey, Active Site Survey, Pre-inspection visits	20
Jobs for Anyone	Virtual Assistant, Local Job, Travel Ready	256

Notes: This table summarizes the main job categories available on Freelancer.com. The second column provides examples of the type of skills workers can list on their profile and employers can link to projects. The third column summarizes the number of individual skills available under each broader job category. See <https://www.freelancer.com/job/> for a full list of skills.

Search Keyword

Online Job

Search

My recent searches

Filter by:

Budget

☐ Fixed Price Projects

min to max

☐ Hourly Projects

min to max

All Durations

☐ Contests

min to max

Type

☐ Local Jobs

☐ Featured Jobs

☐ Recruiter Jobs

☐ Full Time Jobs

Skills

☐ .NET 5.0/6

☐ .NET

☐ PHP

☐ HTML

☐ CSS

Or enter skills

Languages

☐ English

☐ Spanish

☐ German

☐ French

Newest first
 5,000 jobs found, pricing in USD

1 2 3

E-Commerce UI Design Across Platforms 6 days left **\$167** Avg Bid 21 bids

I'm in need of a UI designer who is skilled in creating Corporate and professional-looking web and mobile interfaces for an e-commerce platform. Key responsibilities include: - Developing a user interface compatible with Web, iOS and Android platforms. - Incorporating a sophisticated, corporate and professional design style. - Ensuring that the design promotes user-friendly navigation for online shopping. Ideal candidates should: - Have prior experience in creating UI design across multiple platforms. - Hav.

Graphic Design iPhone Mobile App Development User Interface / IA Website Design

need to made a indicator in esignal charting software 6 days left **\$72 - \$78** 0 bids

I'm looking for a software developer proficient in eSignal and TradingView applications. You will work on creating a specific indicator in eSignal charting software using the logic of a tradingview indicator Key skills and experience: - eSignal charting software - TradingView platform - Understanding of trading and stock market If you possess these skills, this could be the perfect project for you.

C Programming C++ Programming JavaScript Pine Script Python

Instagram Promotion for Brunch/Workshop Event 23 hours left **\$111** Avg Bid 11 bids

I need a savvy Instagram marketer to promote a brunch/workshop I'm hosting. This event is open to both public and private attendees. Your main goal will be to increase event attendance. Ideal skills include: - Experience with Instagram marketing - A knack for engaging event promotion - Understanding of public and private event marketing.

Facebook Marketing Internet Marketing Marketing Social Media Marketing Twitter

URGENT

Bulk Certificate Generation with LaTeX and CSV Integration -- 2 6 days left **\$7** Avg Bid 2 bids

I am in need of a talented freelancer who can assist me in automating the bulk generation of certificates using LaTeX templates and CSV data integration. The project entails developing a streamlined process to efficiently produce certificates for events, workshops, and academic achievements. Key Requirements: 1. LaTeX Proficiency: The freelancer should possess advanced skills in LaTeX typesetting to ensure precise formatting and layout consistency across certificates. 2. CSV Data Integration: Ability to...

Data Entry Excel LaTeX

Amazon, eBay and online Dropshipping Expert 6 days left **\$190** Avg Bid 13 bids

I am currently seeking an experienced dropshipping expert well-versed in product research and sourcing for Amazon and eBay marketplaces. The primary focus of this role will be on electronics. Key Responsibilities: - Research and source high-demand, profitable electronics for the dropshipping business - Analyze market trends and customer reviews to identify potential products - Stay updated with the latest trends in the electronics market Ideal Skills and Experience: - Previous experience in dropshipping...

Figure F.1: Looking for Jobs - Platform Overview

Freelancers For Hire

× Online Users

What work do you require?

Select a category of work

Skills

[Browse skills](#)

Countries

Specific Location

Exams

Hourly Rate (USD)

Rating

★ ★ ★ ★ ★

Online

☒ Online freelancers only (for instant chat)

1 2 3 4 5

Showing 9315 results

•
🇮🇳

mikehurley

🔗 Hire Me

Expert SEO-Backlinks-Local SEO/GMB/Google Ad/SMM

4.9

★★★★★

10.0

\$

6609 reviews

\$10 USD per hour

SEO, Link Building, Internet Marketing, Marketing, Advertising

Highest 6600 ★★★★★ + Reviews, 2000+ satisfied clients Glowing Reviews! Elevate Your Business with Expertise in SEO, Google Ads, and Facebook Ads! # Your digital journey deserves the best! Partner with the most reviewed and skilled SEO, Digital Marketing, and Website Development team of Elite Information Tech for...

•
🇮🇳

seo4quality

🔗 Hire Me

Google #1 Rank | Google Ads | White Hat SEO

4.8

★★★★★

9.6

\$

1564 reviews

\$3 USD per hour

SEO, Internet Marketing, Link Building, Marketing, Search Engine Marketing

★ A Group of Professional Digital Marketing Experts with Google Certified ★ Quick Website SEO Offering the advance level SEO and Google ads service for our all clients to get the good result in terms of both organic and paid way. We have worked on different niche websites. We can perform the best practices for your...

•
🇮🇳

dreamdezine

🔗 Hire Me

Where imagination & art meet.

4.9

★★★★★

9.2

\$

3012 reviews

\$25 USD per hour

Graphic Design, Photoshop, Logo Design, Illustrator, Website Design

Welcome to Dream Dezine, your one-stop destination for top-quality graphic design solutions tailored to elevate your business. Specializing in a diverse array of services including logo design, brochure creation, flyer and poster design, business card development, infographics, eye-catching PowerPoint presentations,...

•
🇮🇳

xyzseo

🔗 Hire Me

SEO|SMM|Google Ads|Wordpress|Shopify|Graphic|Video

4.8

★★★★★

10.0

\$

2652 reviews

\$5 USD per hour

SEO, Internet Marketing, Link Building, Search Engine Marketing, Google Adwords

♦ Digital marketers who make your online brand visibility impressive ♦ White Hat SEO | Google AdWords | Social Media marketing | Web Development We offer complete digital marketing services. We have experience of 12+ years 🔗 Complete Search Engine Optimization (SEO) 🔗 Social Media Optimization (SMO) 🔗 Google...

•
🇮🇳

mhafujmahmud

🔗 Hire Me

Data Entry/Excel/Web Research/Product Listing/VA

5.0

★★★★★

10.0

\$

202 reviews

\$8 USD per hour

Data Entry, Web Search, Research, Search Engine Marketing, Excel

I am a professional freelancer committed to delivering reliable work with 100% accuracy according to my clients' needs. I am detail-oriented and work diligently to meet deadlines. My goal is to offer flexibility, enabling me to explore a variety of jobs that will enhance my skills and share them with future clients....

Figure F.2: Looking for Workers - Platform Overview

Notes: Employers looking for a worker to complete a project can either publicly post a project and open it for competitive bidding, or they can browse through freelancer profiles and directly send a request for hiring to a specific worker. However, the posting of a fixed-price project is the vastly predominant mode by which employers look for workers. The “Freelancers for Hire” section was not a prevalent feature of the platform in 2019-2020.

70

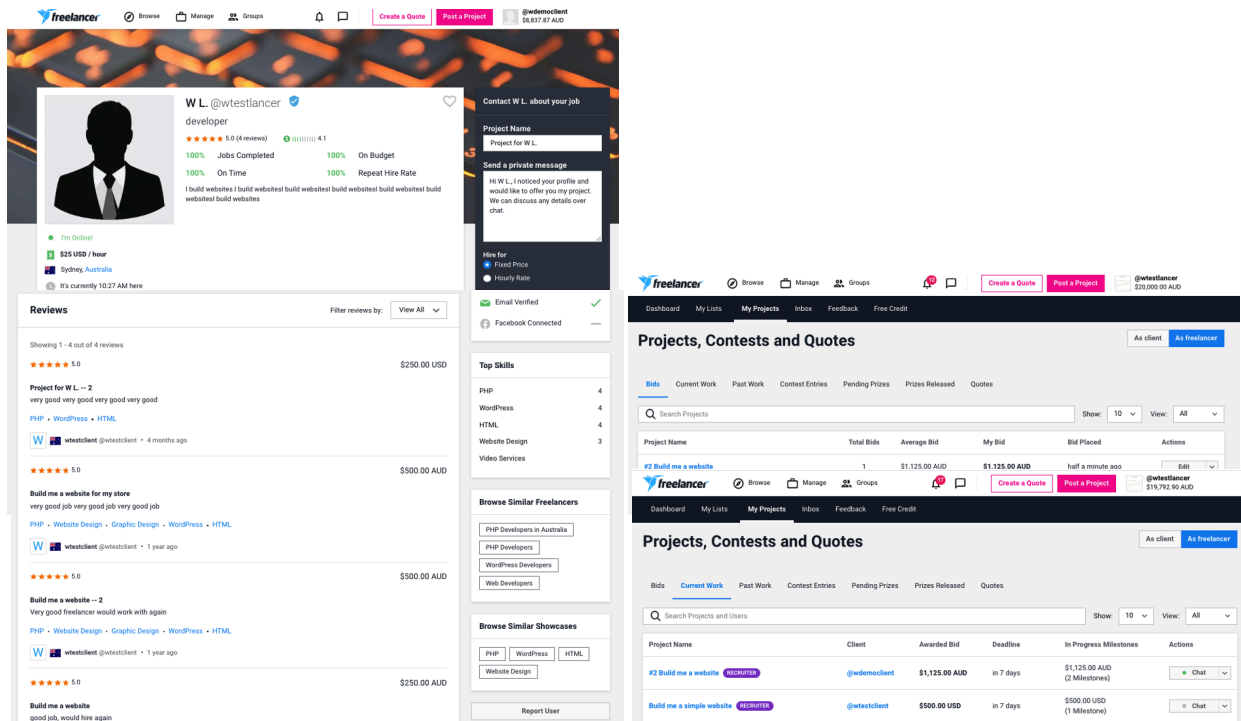


Figure F.3: Worker Profile

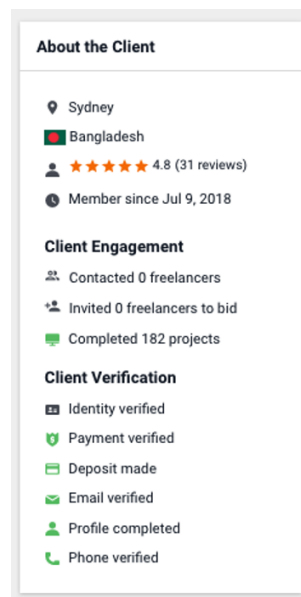


Figure F.4: Employers Profile (from Workers' Point of View)

Try a Freelancer Membership

Designed to maximise your freelancer success and earnings! Save up to 20% on annual plans. Change plans anytime, conditions apply see FAQ.

Monthly Plan
Annual Plan

Basic

\$4.99

per month

Get Started!

50 Bids Per Month

50 Skills

Plus

FREE

~~\$8.95~~

1 month trial

Start Free Trial!

100 Bids Per Month

80 Skills

Professional

\$49.00

per month

Get Started!

300 Bids Per Month

100 Skills

Premier

\$99.00

per month

Get Started!

1500 Bids Per Month

400 Skills

Confirm release of payment for W L.



Build the fronted
\$525.00 AUD

Please ensure that you are satisfied with the work W L. has submitted, as Milestone Payments cannot be returned once released.

☒ My freelancer has delivered satisfactory work.

Cancel
Release Payment

Milestone Payments

Request Milestone

Created Milestones

Date ↕	Description	Status ↕	Amount ↕	
27 Jun 2022	Build the fronted	Release Requested	\$525.00 AUD	Cancel ▼
27 Jun 2022	Build the backend	In Progress	\$600.00 AUD	Request Release ▼

Figure F.5: Memberships and Payments